DESIGN THINKING · ASSIGNMENT 3

IIITA Registration Experience Ideation Stage Report

Mind mapping, group brainstorming and SCAMPER across three problem statements

Group	Group 10
Stage	Ideate
Builds on	Define Stage Report (Assignment 2): problem statements PS-1 and PS-2, five hypotheses, POVs
Course	Design Thinking, BTech V Semester, Section A (July to December 2026)
Submission window	Last week of October 2026

Team

Team member	Roll number	Sticky-note code
Sudhanshu	IIT2024081	SD
Vishwas	IIT2024087	VS
Vichanshu Raj	IIT2024083	VR
Saumya Sood	IIT2024018	SS
Satwik Santosh	IIT2024084	ST
Kyan Mahajan	IIT2024092	KM
Akshat Parmar	IIT2024077	AP
Sarthak Sharma	IIT2024043	SR

Executive Summary

In Assignment 2 we defined two problem statements from our Empathize research on IIITA registration. This stage **diverges** on those two and on the course problem statement, using three methods: individual mind mapping, a timed group brainstorm and SCAMPER. We then **converge** on one concept per problem statement for prototyping.

The three problem statements

Code	Theme	How Might We prompt
PS-T	**Course problem statement**	How might we make public transport (buses and trains) safer, more accessible and more engaging for daily commuters?
PS-1	**Certainty before travel**	How might we help first year students prove, before they leave home, that their registration is complete and accepted?
PS-2	**From gate to a ready room**	How might we guide every first year student from the campus gate to a ready hostel room with confidence at each step?

Requirement checklist

Brief requirement	PS-T	PS-1	PS-2
Mind map: central node plus 3 or more primary branches	4 branches	4 branches	4 branches
Mind map: 2 or more secondary nodes per branch	3 to 4 each (14 nodes)	3 each (12 nodes)	3 each (12 nodes)
Mind map: keywords, colour coding, icons, linked concepts	✔ 3 links	✔ 3 links	✔ 3 links
Brainstorm: 15 or more distinct ideas	18 ideas	18 ideas	18 ideas
Brainstorm: “Yes, and...” on 3 or more ideas	4 builds	3 builds	3 builds
Brainstorm: 3 labelled thematic clusters	✔	✔	✔
SCAMPER: all 7 letters on one existing feature	Stop-request button	Registration advisory PDF	Paper token queue

Report Contents

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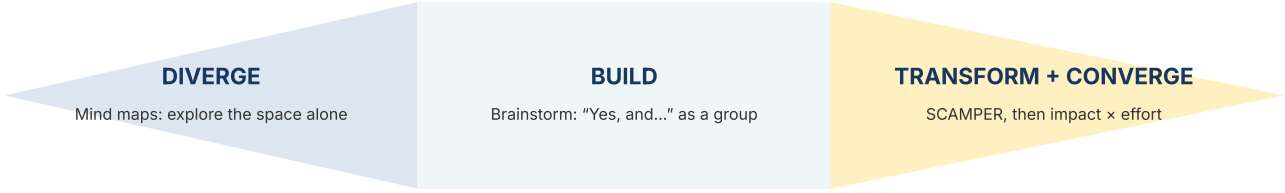
Presenting order

Pages for each speaker, in order.

Speaker	What they present	Pages	Count
1. Saumya Sood IIT2024018	PS-T accessibility map, PS-T brainstorm and team map	4 to 7	4
2. Sudhanshu IIT2024081	PS-T safety map + SCAMPER stop button	8 to 10	3
3. Vishwas IIT2024087	PS-1 ERP / upload map, PS-1 brainstorm and team map	11 to 13	3
4. Vichanshu Raj IIT2024083	PS-1 documents map + SCAMPER advisory PDF	14 to 16	3
5. Satwik Santosh IIT2024084	PS-2 hostel-room map, PS-2 brainstorm and team map	17 to 19	3
6. Kyan Mahajan IIT2024092	PS-2 queue map + SCAMPER paper token	20 to 22	3
7. Akshat Parmar IIT2024077	PS-T engagement map, convergence for all three problems, selected concepts	23 to 27	5
8. Sarthak Sharma IIT2024043	PS-2 family-trust map, Yes-and builds, storyboards, prototype test plan	28 to 33	6

1. Ideation Approach

Each method has a different job. **Mind mapping** is done individually first, so that every member explores the problem space before the group can anchor on one idea. **Brainstorming** uses the group to build quantity and to combine ideas. **SCAMPER** forces structured variation on one feature that already exists, which tends to produce ideas the open brainstorm missed.



Rules we followed

Brainstorm rules (15:00 timer)

- Defer judgement: no "but" for the first 15 minutes
- Go for quantity: target 15 or more ideas per board
- Build on others with **"Yes, and..."**, written as an attached note
- One idea per sticky, with a headline verb
- Each member has their own sticky colour, so contributions stay visible

Staying anchored to evidence

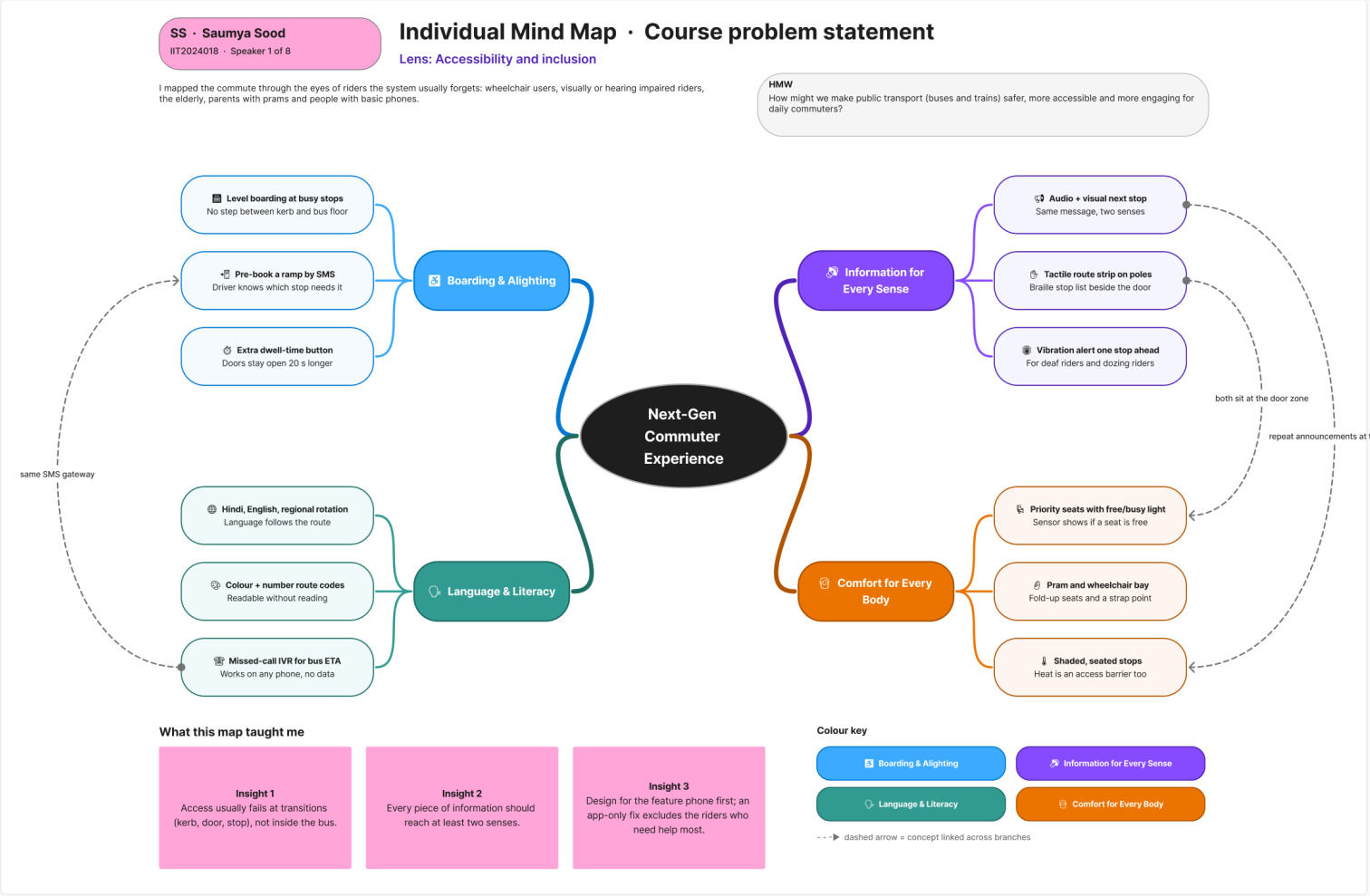
- Every board starts with the HMW from Assignment 2
- The persona POVs (Ananya, Kabir, Rohan, Meera) are pinned beside the board
- Clusters are named after user needs, not technologies
- Ideas are tagged later with the hypothesis they would test (H1 to H5)
- For PS-T, we reused the same lenses: safety, access, engagement

From Define to Ideate

Assignment 2 output	How it feeds this stage
Master POV: students need repeated proof they are on the right path	Central lens for PS-1 and PS-2 mind maps: every branch must produce proof or reassurance
HMW lenses (amplify, eliminate, opposite, assumptions, resources, analogy)	Used as prompts when the brainstorm slowed down; analogies seeded the SCAMPER "Adapt" row
Hypotheses H1 to H5	Convergence criteria: concepts that can test a hypothesis rank higher
Journey-map valleys (verification desk, room entry)	Priority targets for PS-1 and PS-2 respectively

Individual mind map · PS-T · Accessibility and inclusion

How might we make public transport (buses and trains) safer, more accessible and more engaging for daily commuters?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. I mapped the commute through the eyes of riders the system usually forgets: wheelchair users, visually or hearing impaired riders, the elderly, parents with prams and people with basic phones.

HOW TO READ IT

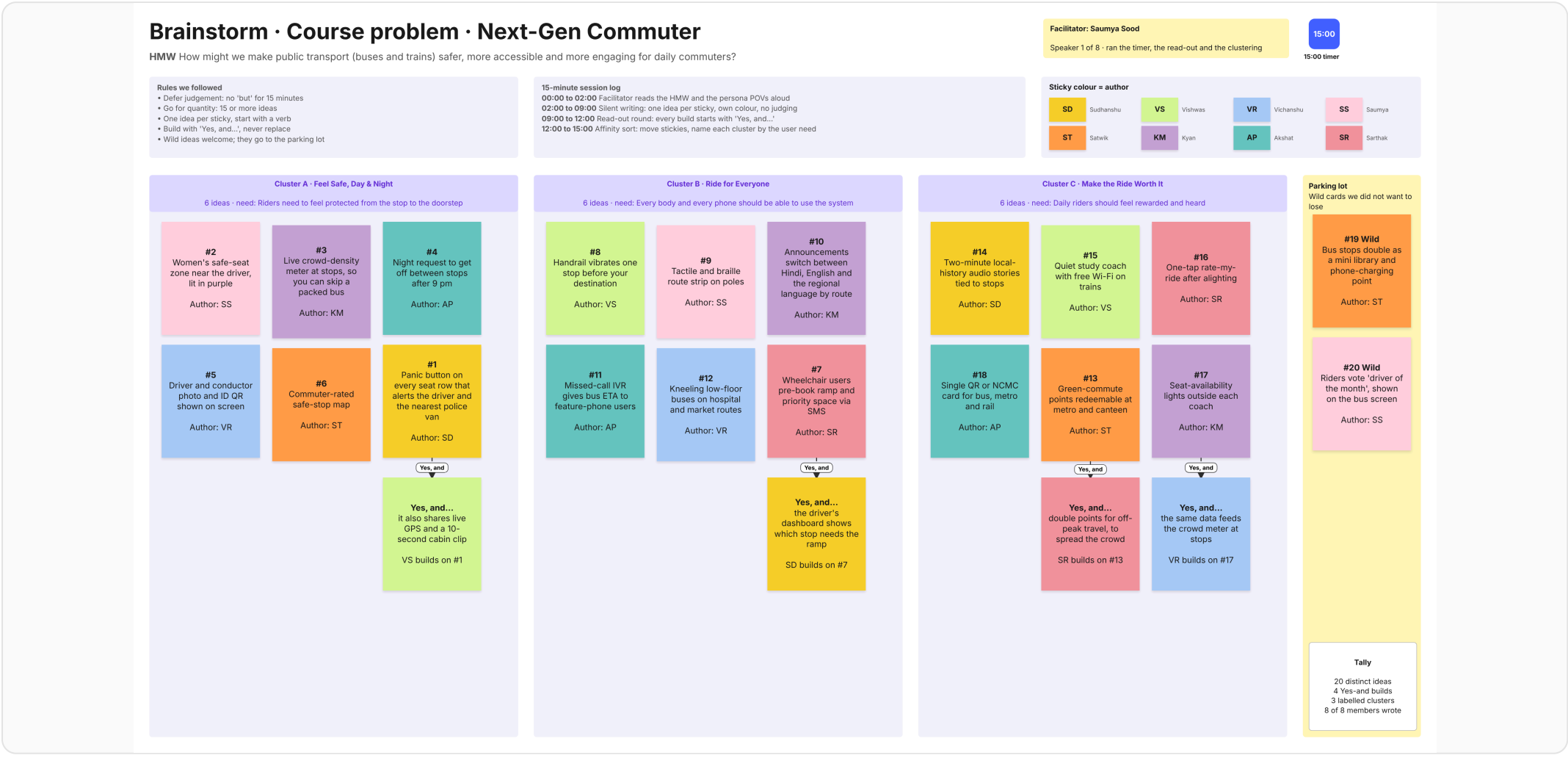
Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

Access usually fails at transitions (kerb, door, stop), not inside the bus. · Every piece of information should reach at least two senses. · Design for the feature phone first; an app-only fix excludes the riders who need help most.

Group brainstorm board (Miro) · PS-T

How might we make public transport (buses and trains) safer, more accessible and more engaging for daily commuters?



WHAT IT IS

Our timed 15-minute team brainstorm on Miro. Everyone wrote ideas on sticky notes (one colour per member) and nobody judged. Saumya Sood facilitated.

HOW TO READ IT

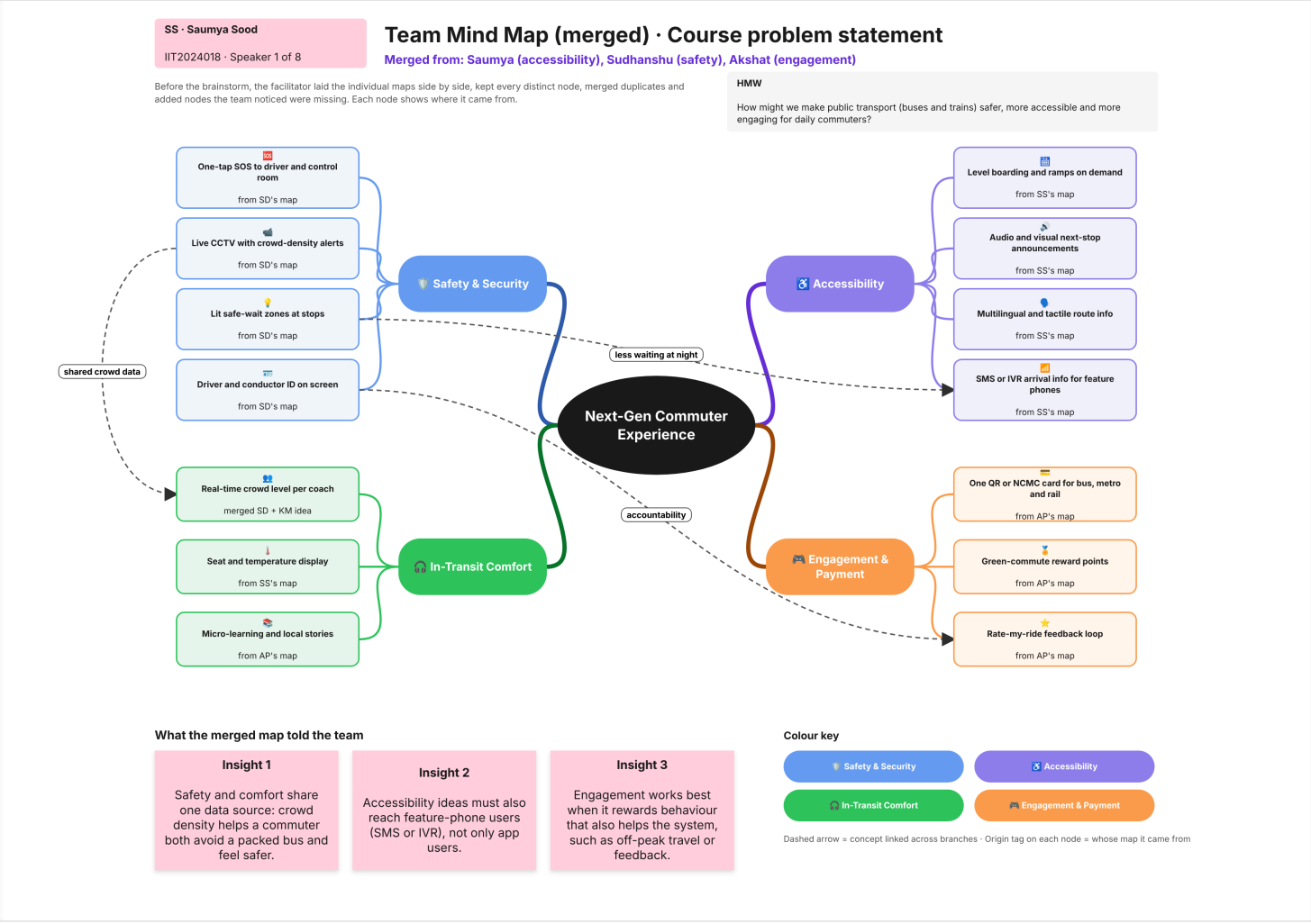
The three tinted areas are the clusters. Arrows labelled 'Yes, and...' show one person building on another's idea. The parking lot holds wild ideas. The side panel has the rules, session log and dot tally.

WHAT IT TELLS US

18 ideas in 3 clusters (Feel Safe, Day & Night, Ride for Everyone, Make the Ride Worth It) plus 2 wild cards, and 4 'Yes, and...' builds. Dot votes on this board went into Akshat's convergence.

Team mind map (merged) · Course problem statement

How might we make public transport (buses and trains) safer, more accessible and more engaging for daily commuters?



WHAT IT IS

Before brainstorming, the facilitator laid the individual maps side by side, kept every distinct node, merged duplicates and added what the team noticed was missing.

HOW TO READ IT

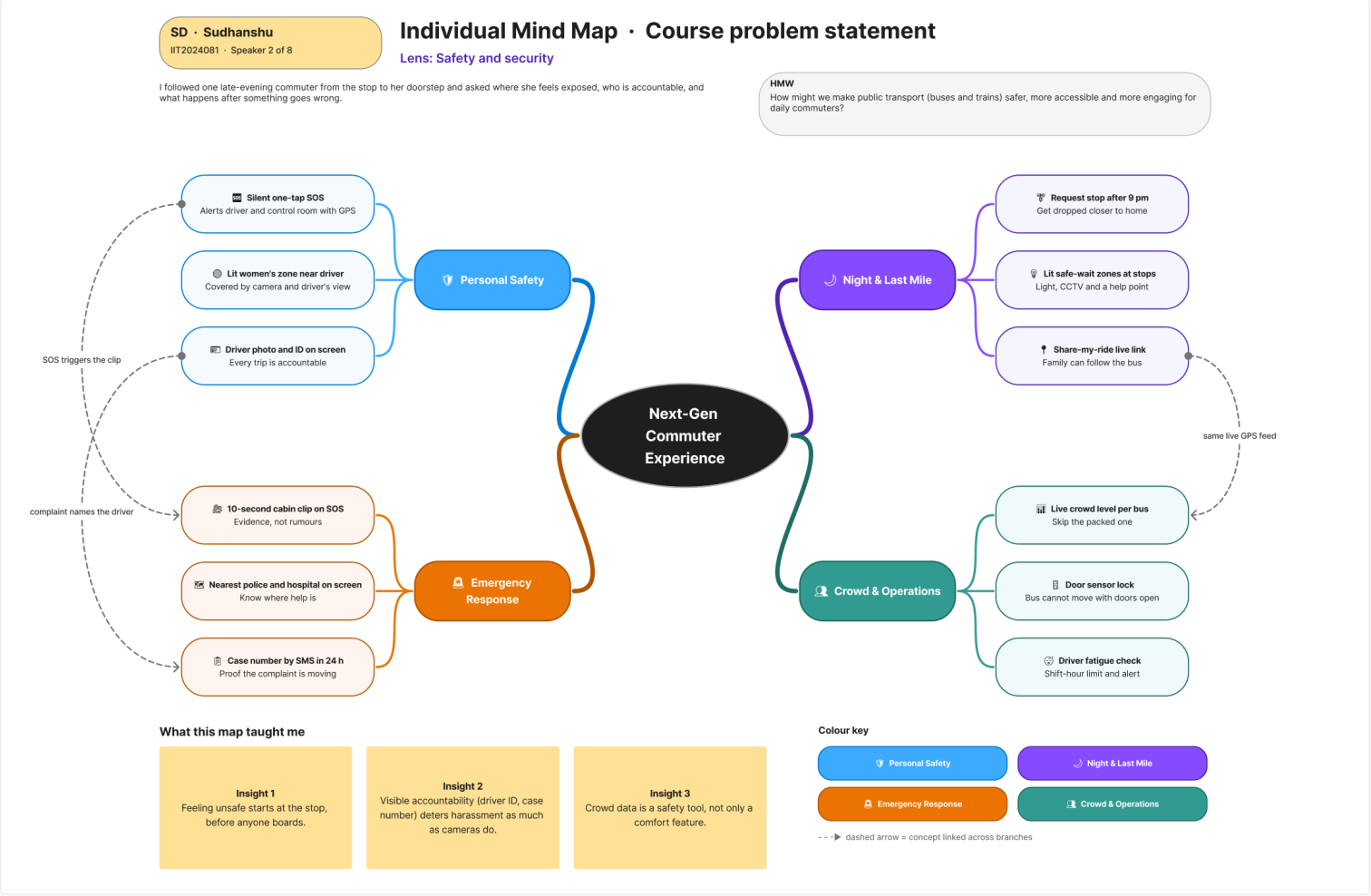
Same colour language as the individual maps. A tag on each node says whose map it came from or that it was added in the merge.

WHAT IT TELLS US

The merged map shows the whole problem space in one picture, so the brainstorm that follows starts wide and not from a single person's view.

Individual mind map · PS-T · Safety and security

How might we make public transport (buses and trains) safer, more accessible and more engaging for daily commuters?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. I followed one late-evening commuter from the stop to her doorstep and asked where she feels exposed, who is accountable, and what happens after something goes wrong.

HOW TO READ IT

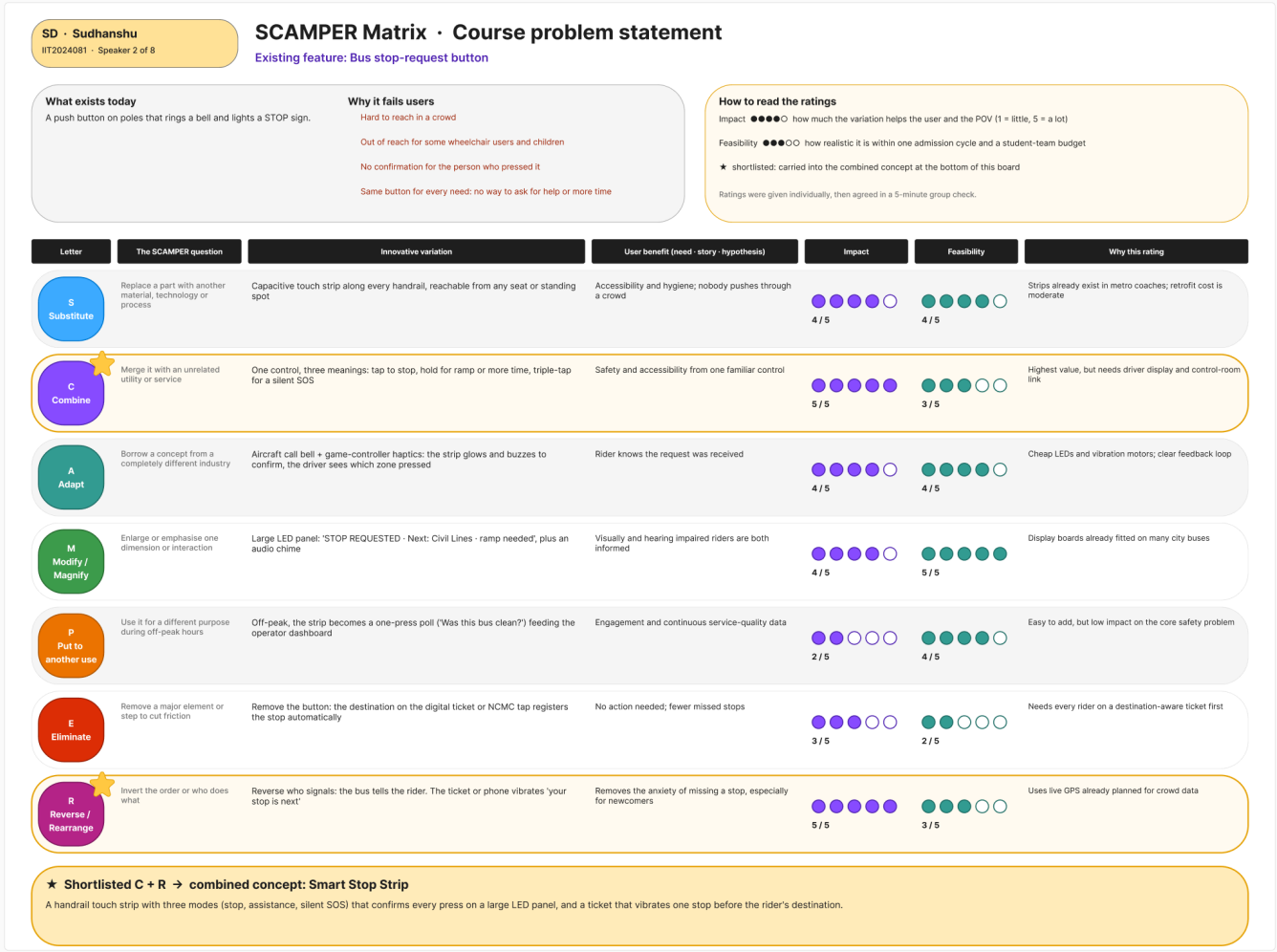
Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

Feeling unsafe starts at the stop, before anyone boards. · Visible accountability (driver ID, case number) deters harassment as much as cameras do. · Crowd data is a safety tool, not only a comfort feature.

SCAMPER matrix · Bus stop-request button

A push button on poles that rings a bell and lights a STOP sign.



SD · Sudhanshu
IIT2024081 · Speaker 2 of 8

SCAMPER Matrix · Course problem statement

Existing feature: Bus stop-request button

What exists today
A push button on poles that rings a bell and lights a STOP sign.

Why it fails users
- Hard to reach in a crowd
- Out of reach for some wheelchair users and children
- No confirmation for the person who pressed it
- Same button for every need: no way to ask for help or more time

How to read the ratings
Impact ●●●●○ how much the variation helps the user and the POV (1 = little, 5 = a lot)
Feasibility ●●●○○ how realistic it is within one admission cycle and a student-team budget
★ shortlisted: carried into the combined concept at the bottom of this board
Ratings were given individually, then agreed in a 5-minute group check.

Letter	The SCAMPER question	Innovative variation	User benefit (need · story · hypothesis)	Impact	Feasibility	Why this rating
S Substitute	Replace a part with another material, technology or process	Capacitive touch strip along every handrail, reachable from any seat or standing spot	Accessibility and hygiene; nobody pushes through a crowd	4 / 5	4 / 5	Strips already exist in metro coaches; retrofit cost is moderate
★ C Combine	Merge it with an unrelated utility or service	One control, three meanings: tap to stop, hold for ramp or more time, triple-tap for a silent SOS	Safety and accessibility from one familiar control	5 / 5	3 / 5	Highest value, but needs driver display and control-room link
A Adapt	Borrow a concept from a completely different industry	Aircraft call bell + game-controller haptics: the strip glows and buzzes to confirm, the driver sees which zone pressed	Rider knows the request was received	4 / 5	4 / 5	Cheap LEDs and vibration motors; clear feedback loop
M Modify / Magnify	Enlarge or emphasise one dimension or interaction	Large LED panel: 'STOP REQUESTED · Next: Civil Lines · ramp needed', plus an audio chime	Visually and hearing impaired riders are both informed	4 / 5	5 / 5	Display boards already fitted on many city buses
P Put to another use	Use it for a different purpose during off-peak hours	Off-peak, the strip becomes a one-press poll ('Was this bus clean?') feeding the operator dashboard	Engagement and continuous service-quality data	2 / 5	4 / 5	Easy to add, but low impact on the core safety problem
E Eliminate	Remove a major element or step to cut friction	Remove the button: the destination on the digital ticket or NCMC tap registers the stop automatically	No action needed; fewer missed stops	3 / 5	2 / 5	Needs every rider on a destination-aware ticket first
★ R Reverse / Rearrange	Invert the order or who does what	Reverse who signals: the bus tells the rider. The ticket or phone vibrates 'your stop is next'	Removes the anxiety of missing a stop, especially for newcomers	5 / 5	3 / 5	Uses live GPS already planned for crowd data

★ Shortlisted C + R → combined concept: Smart Stop Strip
A handrail touch strip with three modes (stop, assistance, silent SOS) that confirms every press on a large LED panel, and a ticket that vibrates one stop before the rider's destination.

WHAT IT IS

SCAMPER is seven prompts (Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse) asked about one existing feature to force fresh variations.

HOW TO READ IT

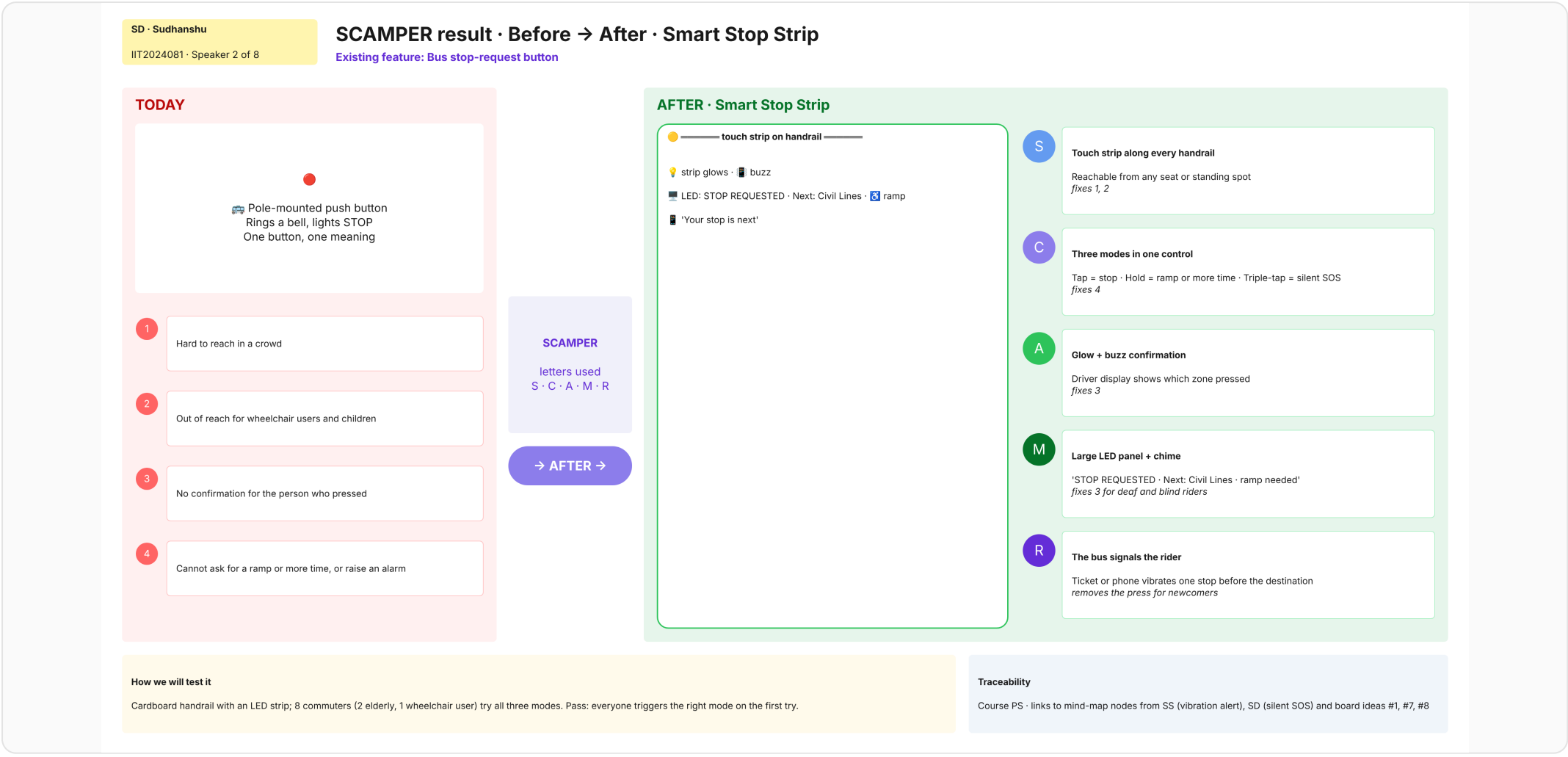
Each row is one letter with the variation, who benefits, and two ratings out of 5 (impact and feasibility). A gold star marks the rows we carried into the concept.

WHAT IT TELLS US

Ratings differ row by row; the stars mark the letters we kept. Result: Smart Stop Strip.

SCAMPER result · before → after · Smart Stop Strip

A push button on poles that rings a bell and lights a STOP sign.



WHAT IT IS

The left side is today's feature with its four pain points; the right side is the concept built from the letters we picked.

HOW TO READ IT

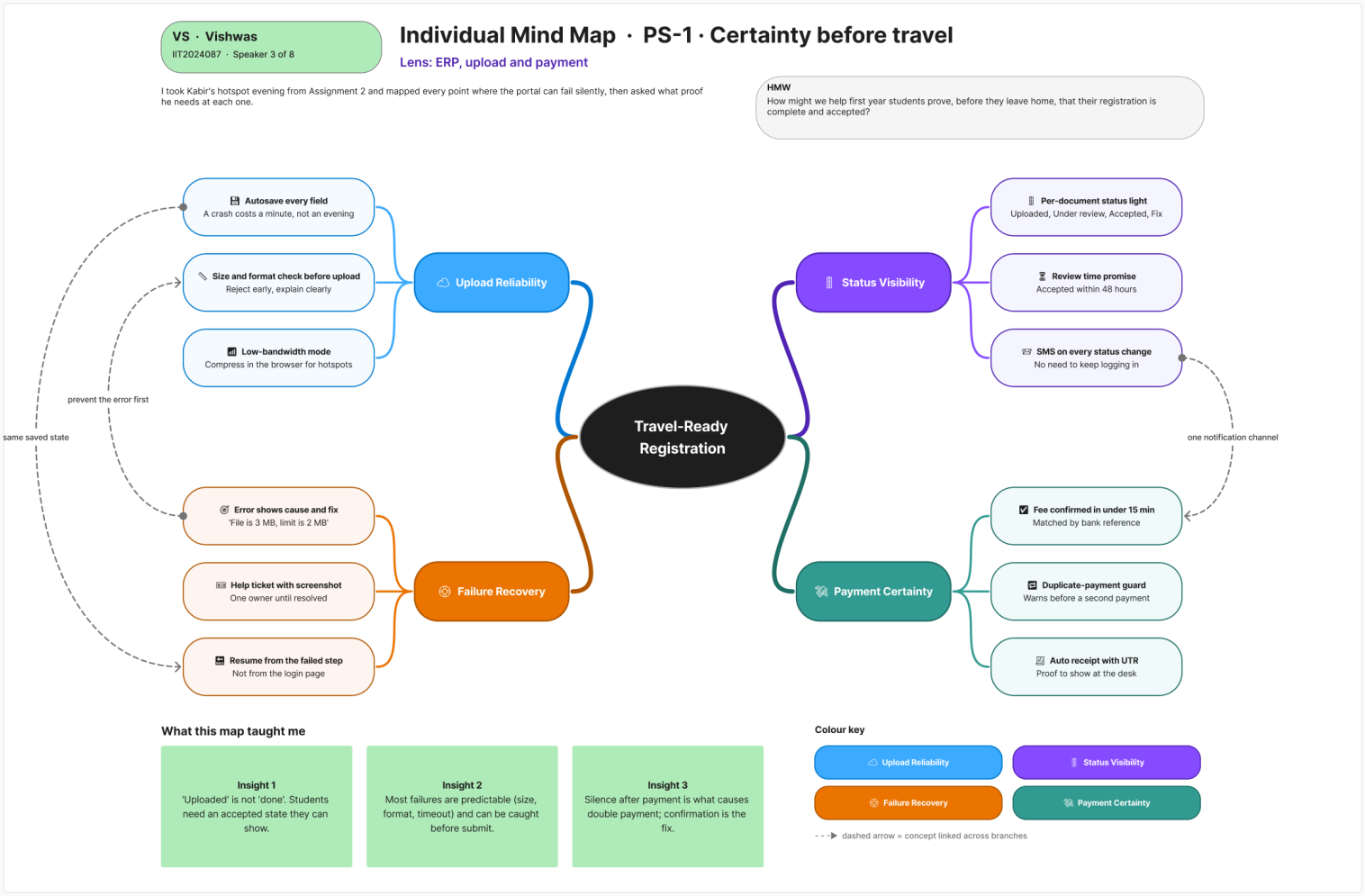
Each coloured circle on the right is the SCAMPER letter that produced that part; the 'fixes' note says which pain it removes. Bottom: how we will test it.

WHAT IT TELLS US

Smart Stop Strip answers every pain on the left and comes with a cheap prototype test and a pass mark.

Individual mind map · PS-1 · ERP, upload and payment

How might we help first year students prove, before they leave home, that their registration is complete and accepted?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. I took Kabir's hotspot evening from Assignment 2 and mapped every point where the portal can fail silently, then asked what proof he needs at each one.

HOW TO READ IT

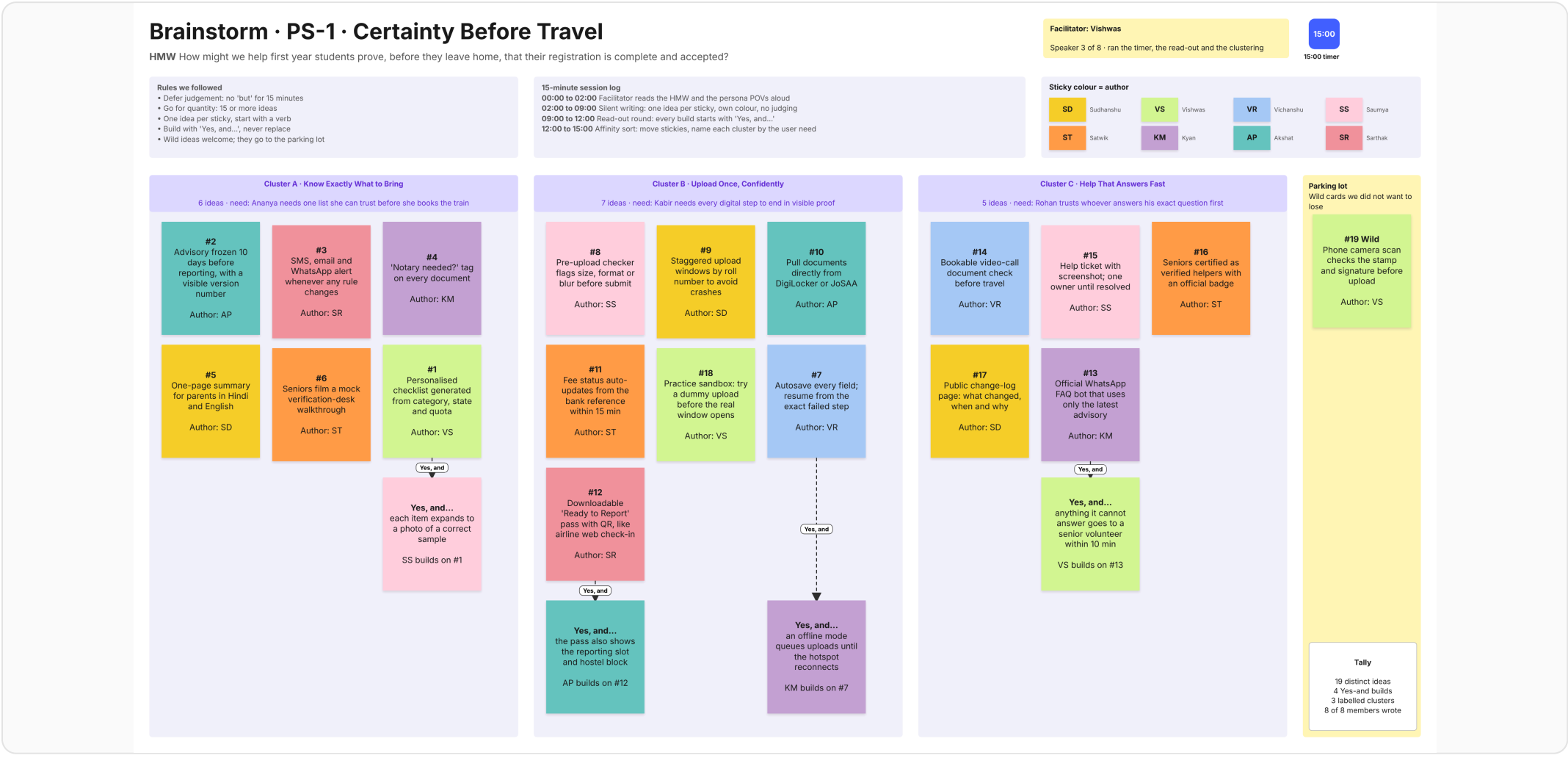
Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

'Uploaded' is not 'done'. Students need an accepted state they can show. · Most failures are predictable (size, format, timeout) and can be caught before submit. · Silence after payment is what causes double payment; confirmation is the fix.

Group brainstorm board (Miro) · PS-1

How might we help first year students prove, before they leave home, that their registration is complete and accepted?



WHAT IT IS

Our timed 15-minute team brainstorm on Miro. Everyone wrote ideas on sticky notes (one colour per member) and nobody judged. Vishwas facilitated.

HOW TO READ IT

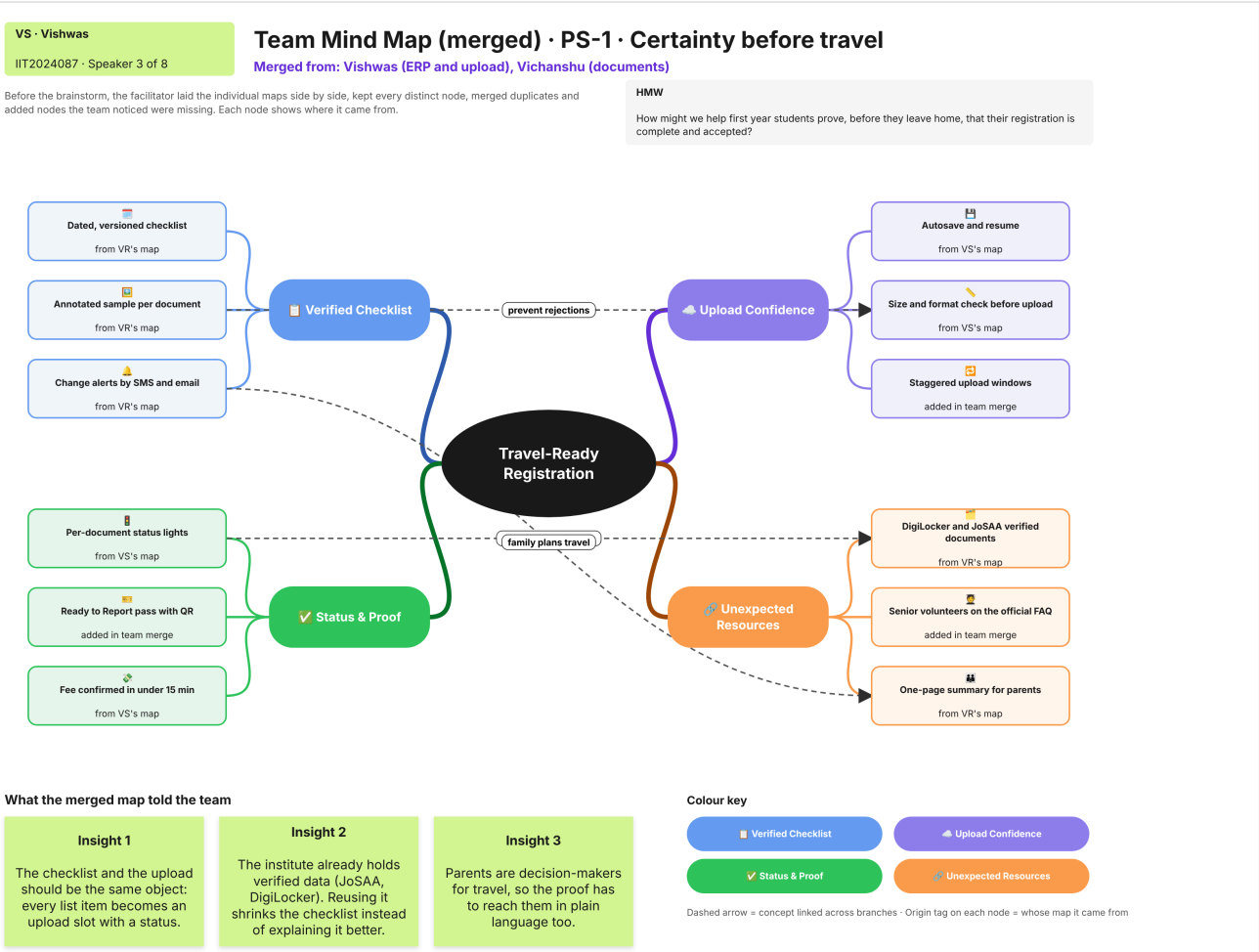
The three tinted areas are the clusters. Arrows labelled 'Yes, and...' show one person building on another's idea. The parking lot holds wild ideas. The side panel has the rules, session log and dot tally.

WHAT IT TELLS US

18 ideas in 3 clusters (Know Exactly What to Bring, Upload Once, Confidently, Help That Answers Fast) plus 1 wild card, and 4 'Yes, and...' builds. Dot votes on this board went into Akshat's convergence.

Team mind map (merged) · PS-1 · Certainty before travel

How might we help first year students prove, before they leave home, that their registration is complete and accepted?



WHAT IT IS

Before brainstorming, the facilitator laid the individual maps side by side, kept every distinct node, merged duplicates and added what the team noticed was missing.

HOW TO READ IT

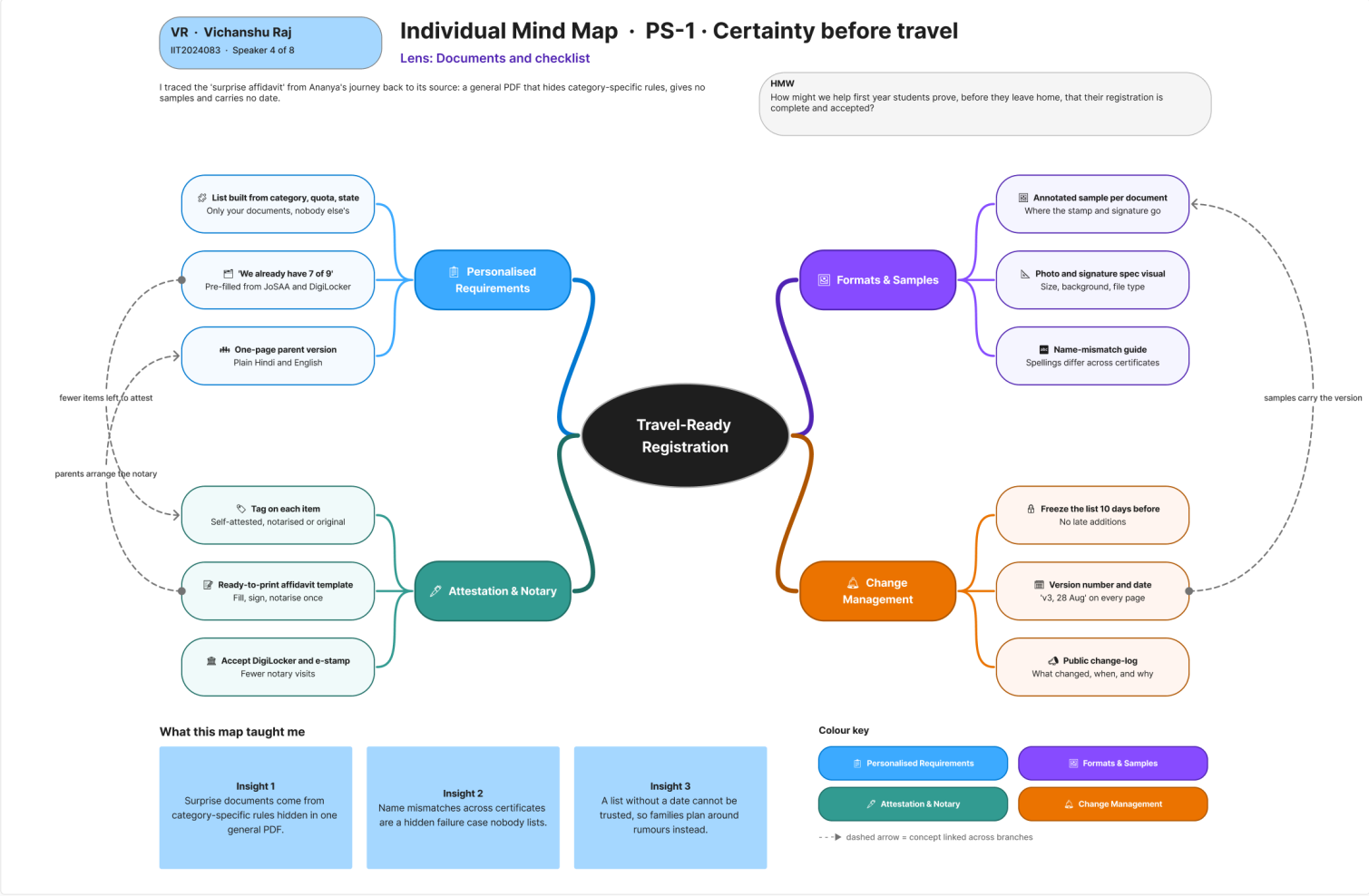
Same colour language as the individual maps. A tag on each node says whose map it came from or that it was added in the merge.

WHAT IT TELLS US

The merged map shows the whole problem space in one picture, so the brainstorm that follows starts wide and not from a single person's view.

Individual mind map · PS-1 · Documents and checklist

How might we help first year students prove, before they leave home, that their registration is complete and accepted?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. I traced the 'surprise affidavit' from Ananya's journey back to its source: a general PDF that hides category-specific rules, gives no samples and carries no date.

HOW TO READ IT

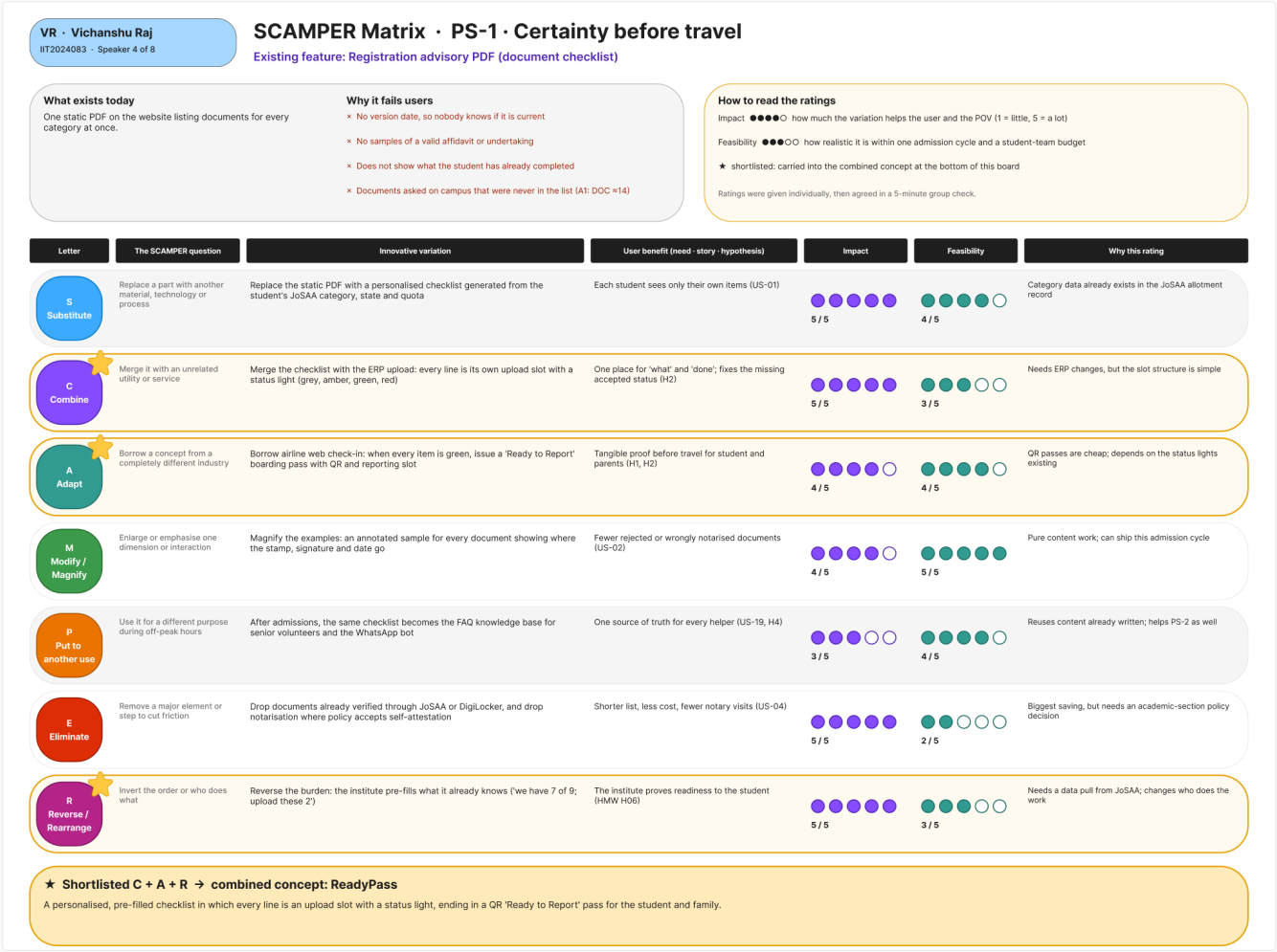
Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

Surprise documents come from category-specific rules hidden in one general PDF. · Name mismatches across certificates are a hidden failure case nobody lists. · A list without a date cannot be trusted, so families plan around rumours instead.

SCAMPER matrix · Registration advisory PDF (document checklist)

One static PDF on the website listing documents for every category at once.



VR · Vichanshu Raj
IIT2024083 · Speaker 4 of 8

SCAMPER Matrix · PS-1 · Certainty before travel

Existing feature: Registration advisory PDF (document checklist)

What exists today
One static PDF on the website listing documents for every category at once.

Why it fails users
- × No version date, so nobody knows if it is current
- × No samples of a valid affidavit or undertaking
- × Does not show what the student has already completed
- × Documents asked on campus that were never in the list (A1: DOC ≈14)

How to read the ratings
Impact ●●●●○ how much the variation helps the user and the POV (1 = little, 5 = a lot)
Feasibility ●●●○○ how realistic it is within one admission cycle and a student-team budget
★ shortlisted: carried into the combined concept at the bottom of this board
Ratings were given individually, then agreed in a 5-minute group check.

Letter	The SCAMPER question	Innovative variation	User benefit (need · story · hypothesis)	Impact	Feasibility	Why this rating
S Substitute	Replace a part with another material, technology or process	Replace the static PDF with a personalised checklist generated from the student's JoSAA category, state and quota	Each student sees only their own items (US-01)	5 / 5	4 / 5	Category data already exists in the JoSAA allotment record
★ C Combine	Merge it with an unrelated utility or service	Merge the checklist with the ERP upload: every line is its own upload slot with a status light (grey, amber, green, red)	One place for 'what' and 'done'; fixes the missing accepted status (H2)	5 / 5	3 / 5	Needs ERP changes, but the slot structure is simple
★ A Adapt	Borrow a concept from a completely different industry	Borrow airline web check-in: when every item is green, issue a 'Ready to Report' boarding pass with QR and reporting slot	Tangible proof before travel for student and parents (H1, H2)	4 / 5	4 / 5	QR passes are cheap; depends on the status lights existing
M Modify / Magnify	Enlarge or emphasise one dimension or interaction	Magnify the examples: an annotated sample for every document showing where the stamp, signature and date go	Fewer rejected or wrongly notarised documents (US-02)	4 / 5	5 / 5	Pure content work; can ship this admission cycle
P Put to another use	Use it for a different purpose during off-peak hours	After admissions, the same checklist becomes the FAQ knowledge base for senior volunteers and the WhatsApp bot	One source of truth for every helper (US-19, H4)	3 / 5	4 / 5	Reuses content already written; helps PS-2 as well
E Eliminate	Remove a major element or step to cut friction	Drop documents already verified through JoSAA or DigiLocker, and drop notarisation where policy accepts self-attestation	Shorter list, less cost, fewer notary visits (US-04)	5 / 5	2 / 5	Biggest saving, but needs an academic-section policy decision
★ R Reverse / Rearrange	Invert the order or who does what	Reverse the burden: the institute pre-fills what it already knows ('we have 7 of 9; upload these 2')	The institute proves readiness to the student (HMW H06)	5 / 5	3 / 5	Needs a data pull from JoSAA; changes who does the work

★ Shortlisted C + A + R → combined concept: ReadyPass
A personalised, pre-filled checklist in which every line is an upload slot with a status light, ending in a QR 'Ready to Report' pass for the student and family.

WHAT IT IS

SCAMPER is seven prompts (Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse) asked about one existing feature to force fresh variations.

HOW TO READ IT

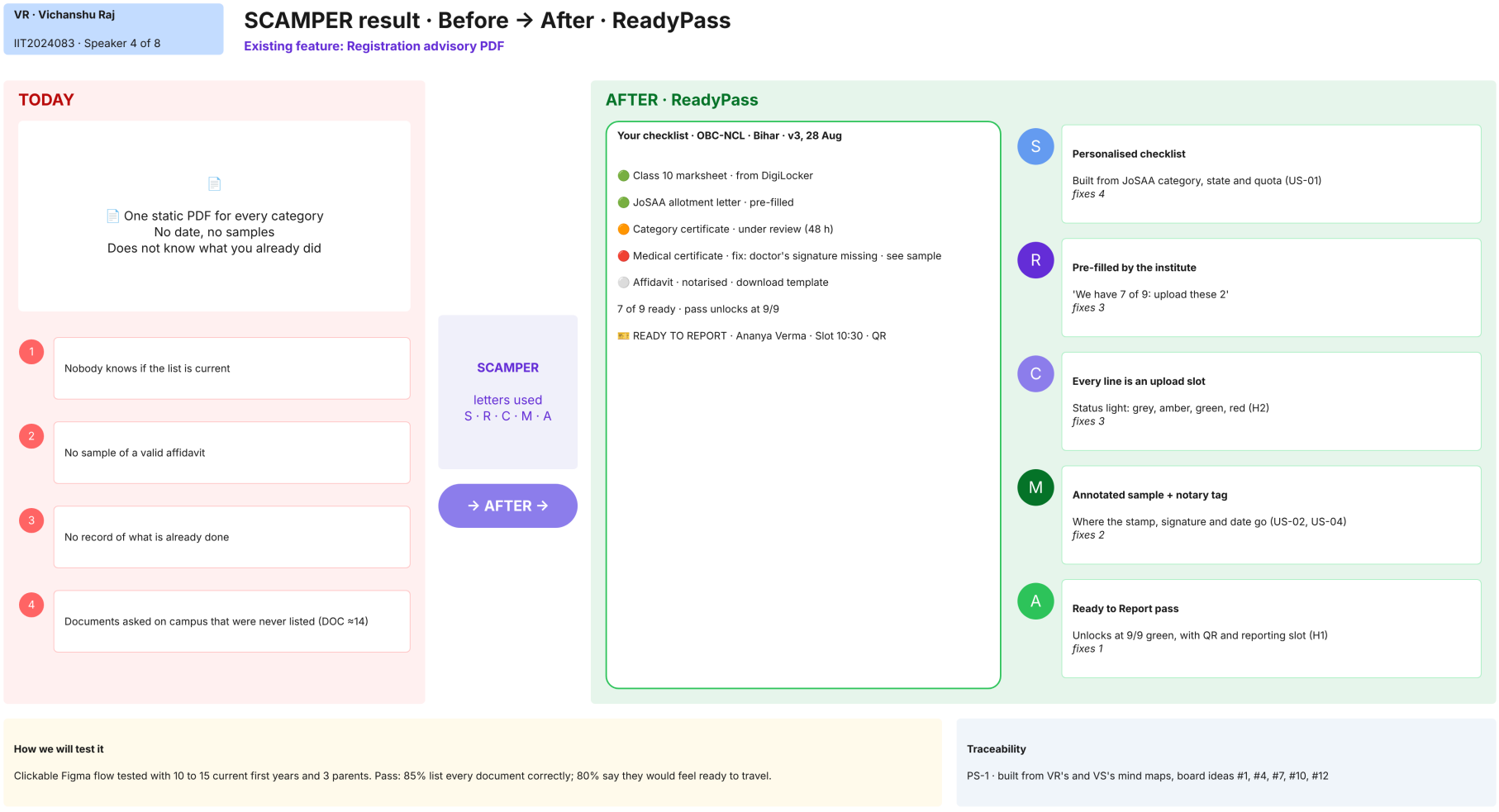
Each row is one letter with the variation, who benefits, and two ratings out of 5 (impact and feasibility). A gold star marks the rows we carried into the concept.

WHAT IT TELLS US

Ratings differ row by row; the stars mark the letters we kept. Result: ReadyPass.

SCAMPER result · before → after · ReadyPass

One static PDF on the website listing documents for every category at once.



WHAT IT IS

The left side is today's feature with its four pain points; the right side is the concept built from the letters we picked.

HOW TO READ IT

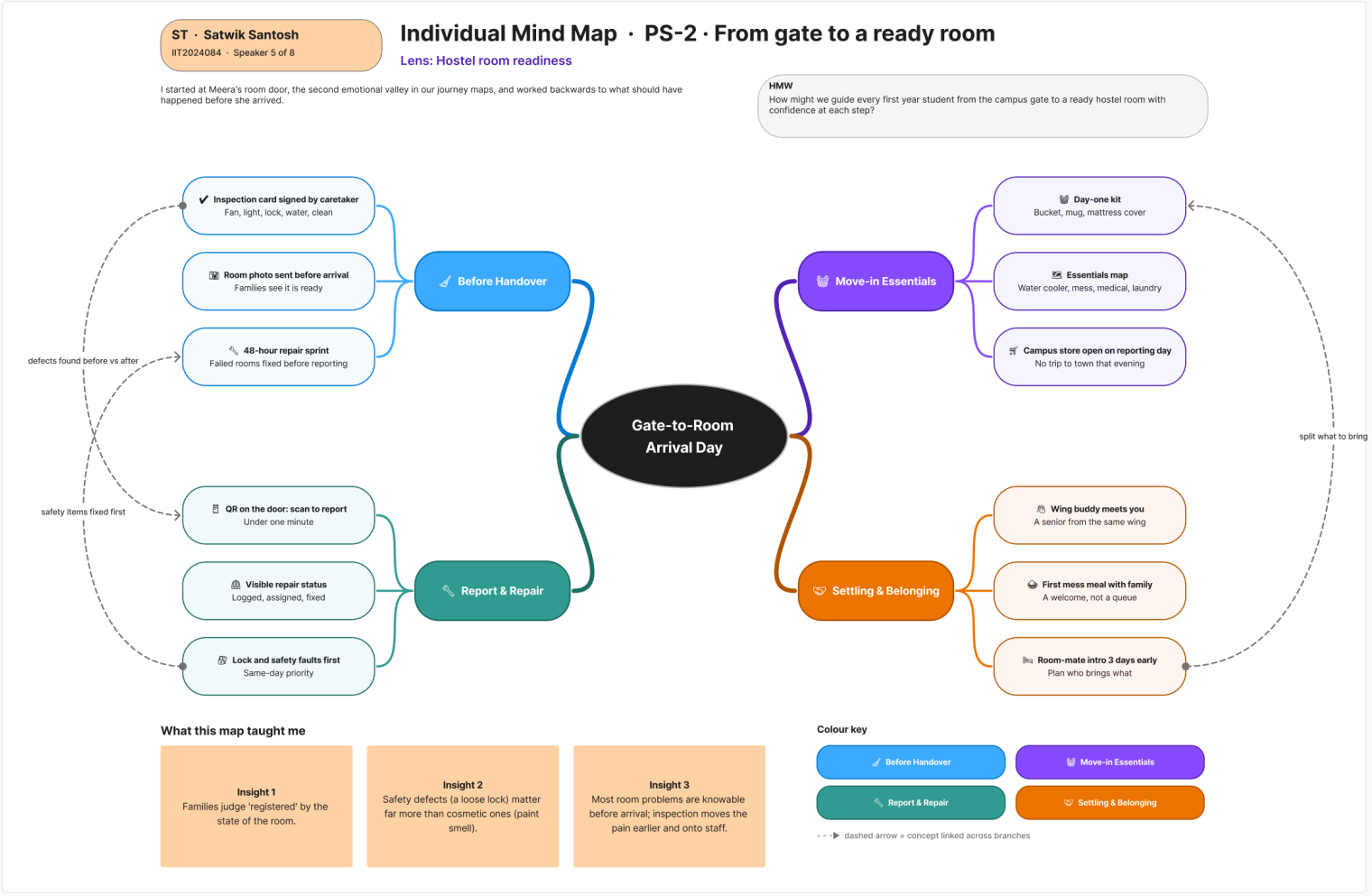
Each coloured circle on the right is the SCAMPER letter that produced that part; the 'fixes' note says which pain it removes. Bottom: how we will test it.

WHAT IT TELLS US

ReadyPass answers every pain on the left and comes with a cheap prototype test and a pass mark.

Individual mind map · PS-2 · Hostel room readiness

How might we guide every first year student from the campus gate to a ready hostel room with confidence at each step?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. I started at Meera's room door, the second emotional valley in our journey maps, and worked backwards to what should have happened before she arrived.

HOW TO READ IT

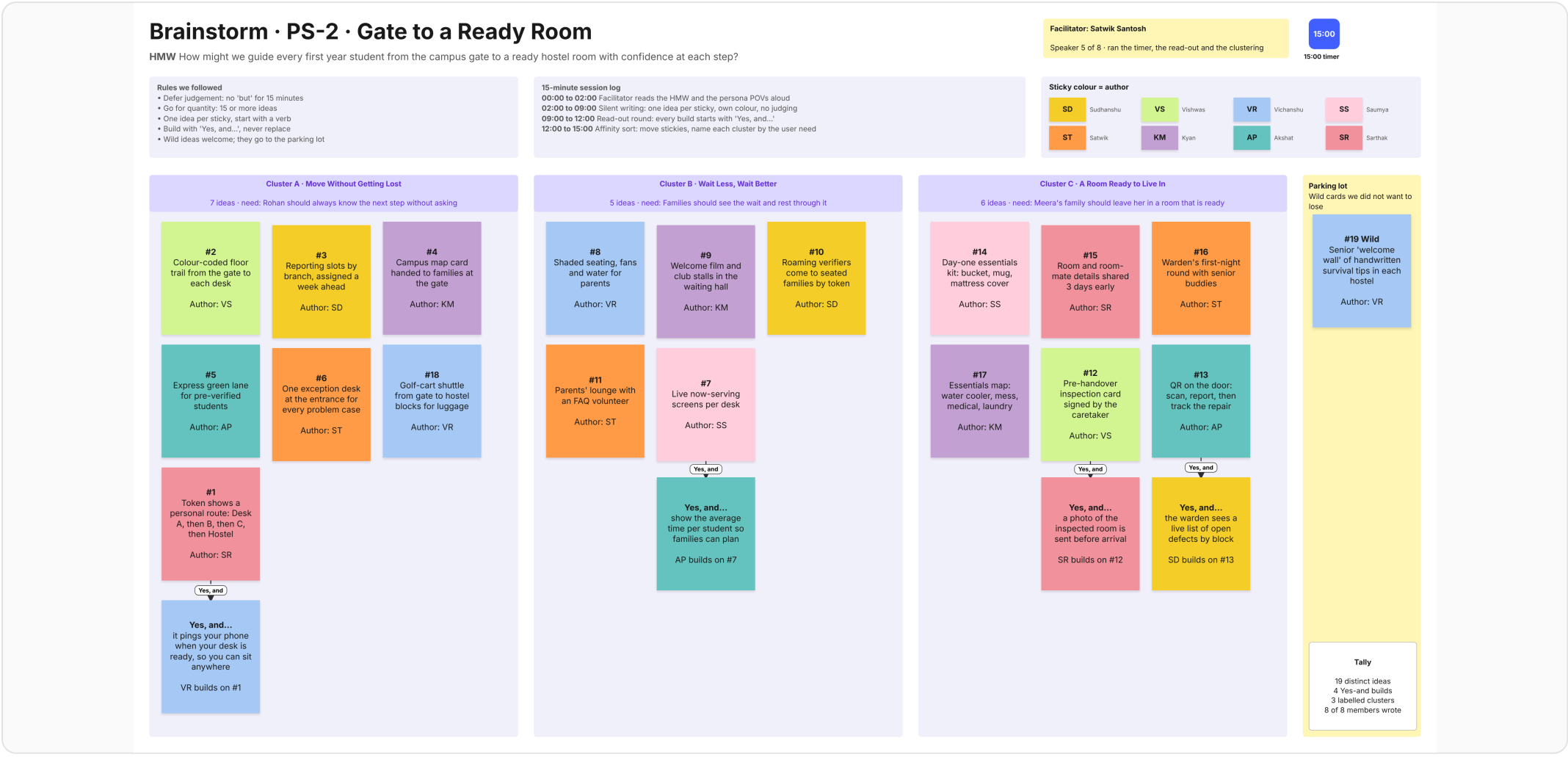
Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

Families judge 'registered' by the state of the room. · Safety defects (a loose lock) matter far more than cosmetic ones (paint smell). · Most room problems are knowable before arrival; inspection moves the pain earlier and onto staff.

Group brainstorm board (Miro) · PS-2

How might we guide every first year student from the campus gate to a ready hostel room with confidence at each step?



WHAT IT IS

Our timed 15-minute team brainstorm on Miro. Everyone wrote ideas on sticky notes (one colour per member) and nobody judged. Satwik Santosh facilitated.

HOW TO READ IT

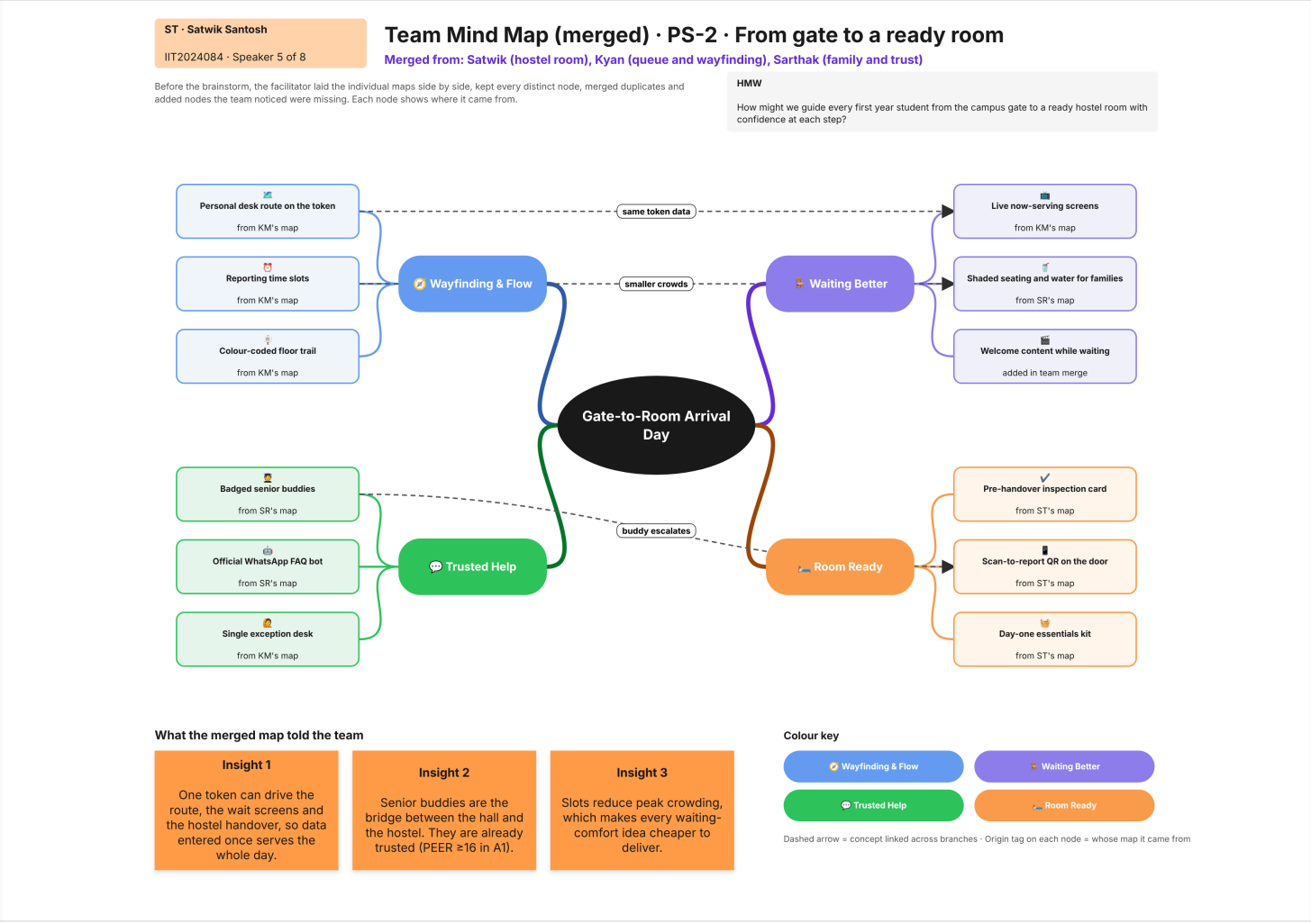
The three tinted areas are the clusters. Arrows labelled 'Yes, and...' show one person building on another's idea. The parking lot holds wild ideas. The side panel has the rules, session log and dot tally.

WHAT IT TELLS US

18 ideas in 3 clusters (Move Without Getting Lost, Wait Less, Wait Better, A Room Ready to Live In) plus 1 wild card, and 4 'Yes, and...' builds. Dot votes on this board went into Akshat's convergence.

Team mind map (merged) · PS-2 · From gate to a ready room

How might we guide every first year student from the campus gate to a ready hostel room with confidence at each step?



WHAT IT IS

Before brainstorming, the facilitator laid the individual maps side by side, kept every distinct node, merged duplicates and added what the team noticed was missing.

HOW TO READ IT

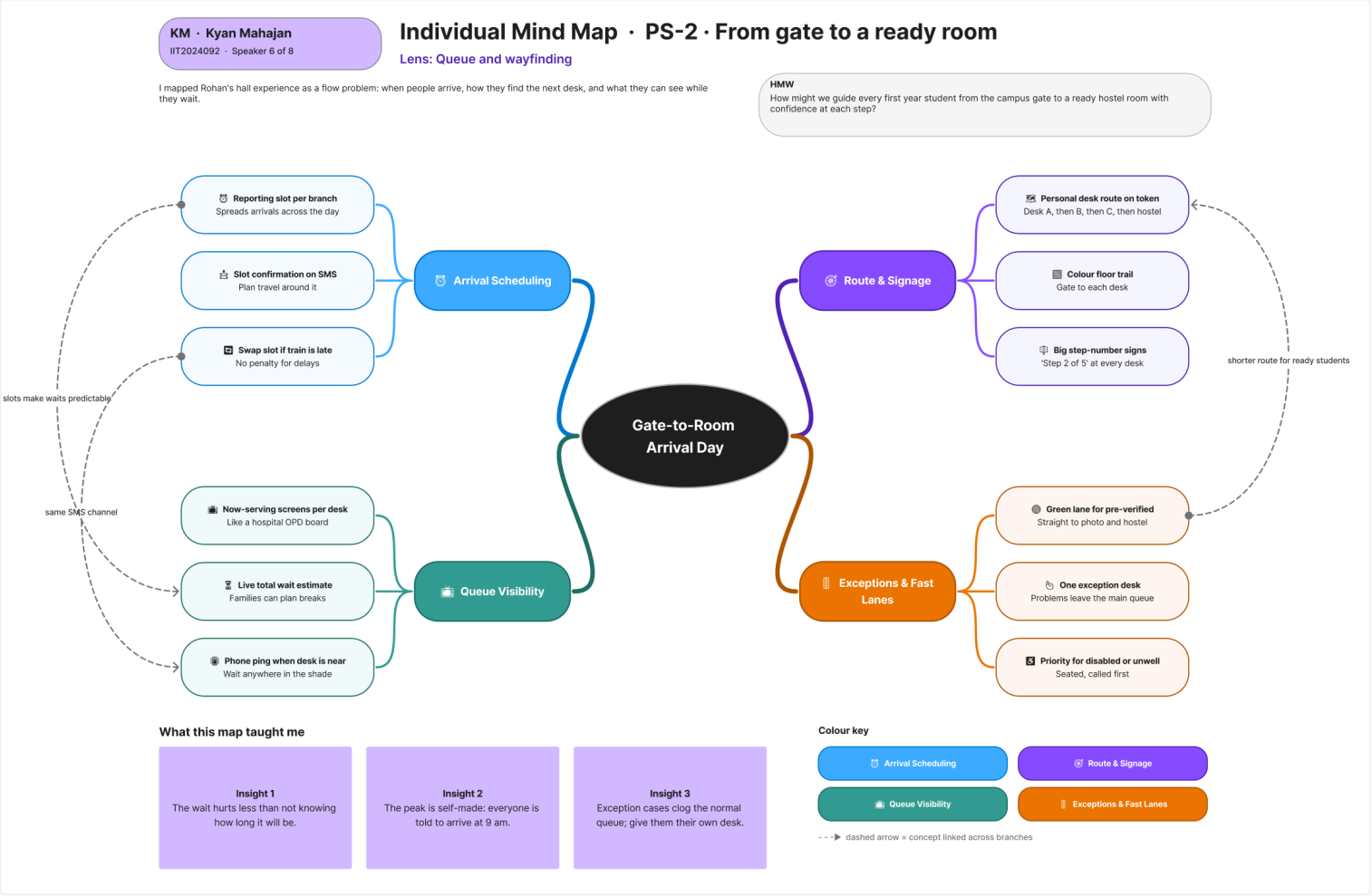
Same colour language as the individual maps. A tag on each node says whose map it came from or that it was added in the merge.

WHAT IT TELLS US

The merged map shows the whole problem space in one picture, so the brainstorm that follows starts wide and not from a single person's view.

Individual mind map · PS-2 · Queue and wayfinding

How might we guide every first year student from the campus gate to a ready hostel room with confidence at each step?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. I mapped Rohan's hall experience as a flow problem: when people arrive, how they find the next desk, and what they can see while they wait.

HOW TO READ IT

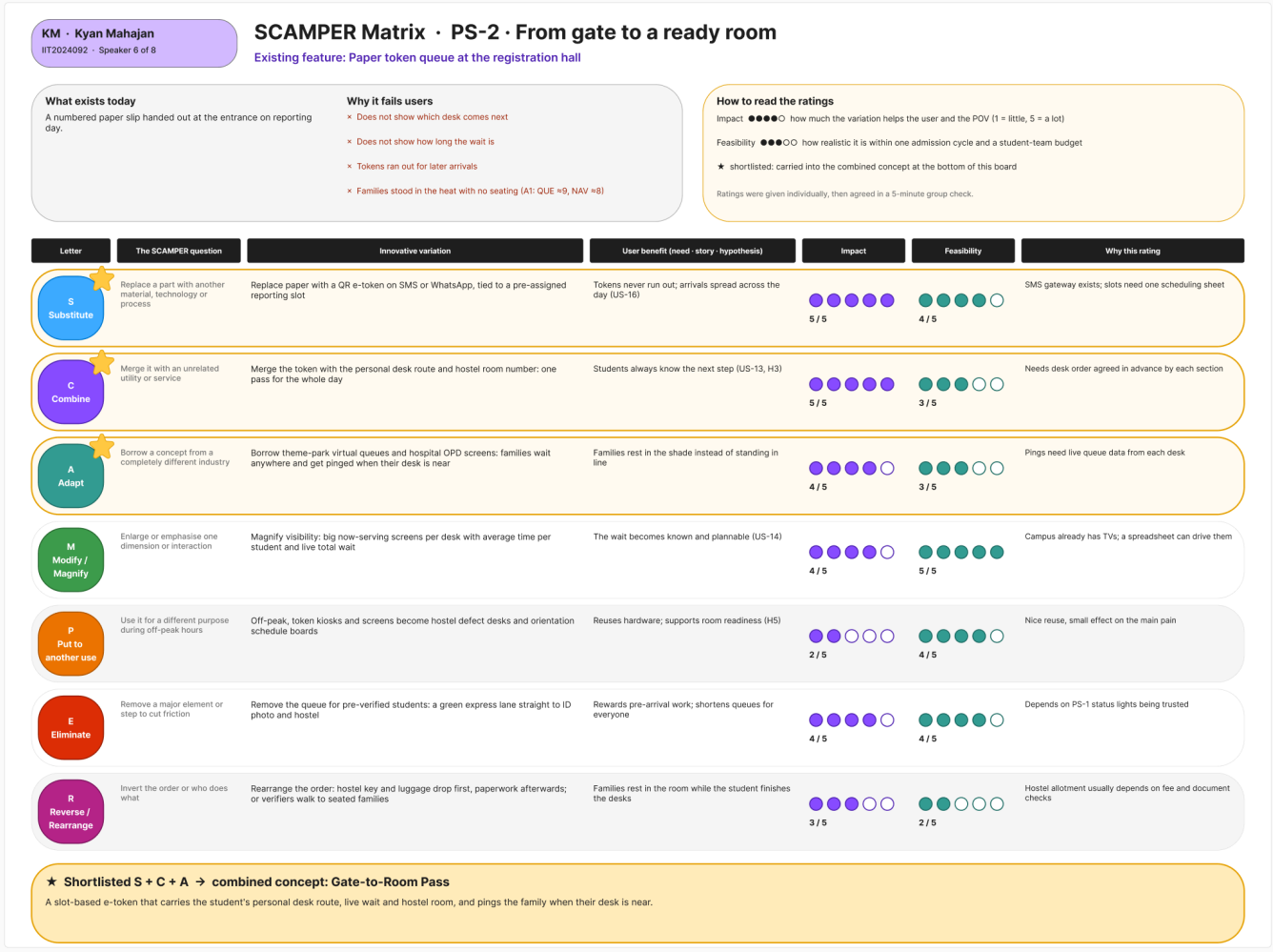
Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

The wait hurts less than not knowing how long it will be. · The peak is self-made: everyone is told to arrive at 9 am. · Exception cases clog the normal queue; give them their own desk.

SCAMPER matrix · Paper token queue at the registration hall

A numbered paper slip handed out at the entrance on reporting day.



KM · Kyan Mahajan
IIT2024092 · Speaker 6 of 8

SCAMPER Matrix · PS-2 · From gate to a ready room

Existing feature: Paper token queue at the registration hall

What exists today

A numbered paper slip handed out at the entrance on reporting day.

Why it fails users

- × Does not show which desk comes next
- × Does not show how long the wait is
- × Tokens ran out for later arrivals
- × Families stood in the heat with no seating (A1: QUE ≈9, NAV ≈8)

How to read the ratings

Impact ●●●●○ how much the variation helps the user and the POV (1 = little, 5 = a lot)

Feasibility ●●●○○ how realistic it is within one admission cycle and a student-team budget

★ shortlisted: carried into the combined concept at the bottom of this board

Ratings were given individually, then agreed in a 5-minute group check.

Letter	The SCAMPER question	Innovative variation	User benefit (need · story · hypothesis)	Impact	Feasibility	Why this rating
★ S Substitute	Replace a part with another material, technology or process	Replace paper with a QR e-token on SMS or WhatsApp, tied to a pre-assigned reporting slot	Tokens never run out; arrivals spread across the day (US-16)	5 / 5	4 / 5	SMS gateway exists; slots need one scheduling sheet
★ C Combine	Merge it with an unrelated utility or service	Merge the token with the personal desk route and hostel room number: one pass for the whole day	Students always know the next step (US-13, H3)	5 / 5	3 / 5	Needs desk order agreed in advance by each section
★ A Adapt	Borrow a concept from a completely different industry	Borrow theme-park virtual queues and hospital OPD screens: families wait anywhere and get pinged when their desk is near	Families rest in the shade instead of standing in line	4 / 5	3 / 5	Pings need live queue data from each desk
M Modify / Magnify	Enlarge or emphasise one dimension or interaction	Magnify visibility: big now-serving screens per desk with average time per student and live total wait	The wait becomes known and plannable (US-14)	4 / 5	5 / 5	Campus already has TVs; a spreadsheet can drive them
P Put to another use	Use it for a different purpose during off-peak hours	Off-peak, token kiosks and screens become hostel defect desks and orientation schedule boards	Reuses hardware; supports room readiness (H5)	2 / 5	4 / 5	Nice reuse, small effect on the main pain
E Eliminate	Remove a major element or step to cut friction	Remove the queue for pre-verified students: a green express lane straight to ID photo and hostel	Rewards pre-arrival work; shortens queues for everyone	4 / 5	4 / 5	Depends on PS-1 status lights being trusted
R Reverse / Rearrange	Invert the order or who does what	Rearrange the order: hostel key and luggage drop first, paperwork afterwards; or verifiers walk to seated families	Families rest in the room while the student finishes the desks	3 / 5	2 / 5	Hostel allotment usually depends on fee and document checks

★ Shortlisted S + C + A → combined concept: Gate-to-Room Pass

A slot-based e-token that carries the student's personal desk route, live wait and hostel room, and pings the family when their desk is near.

WHAT IT IS

SCAMPER is seven prompts (Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, Reverse) asked about one existing feature to force fresh variations.

HOW TO READ IT

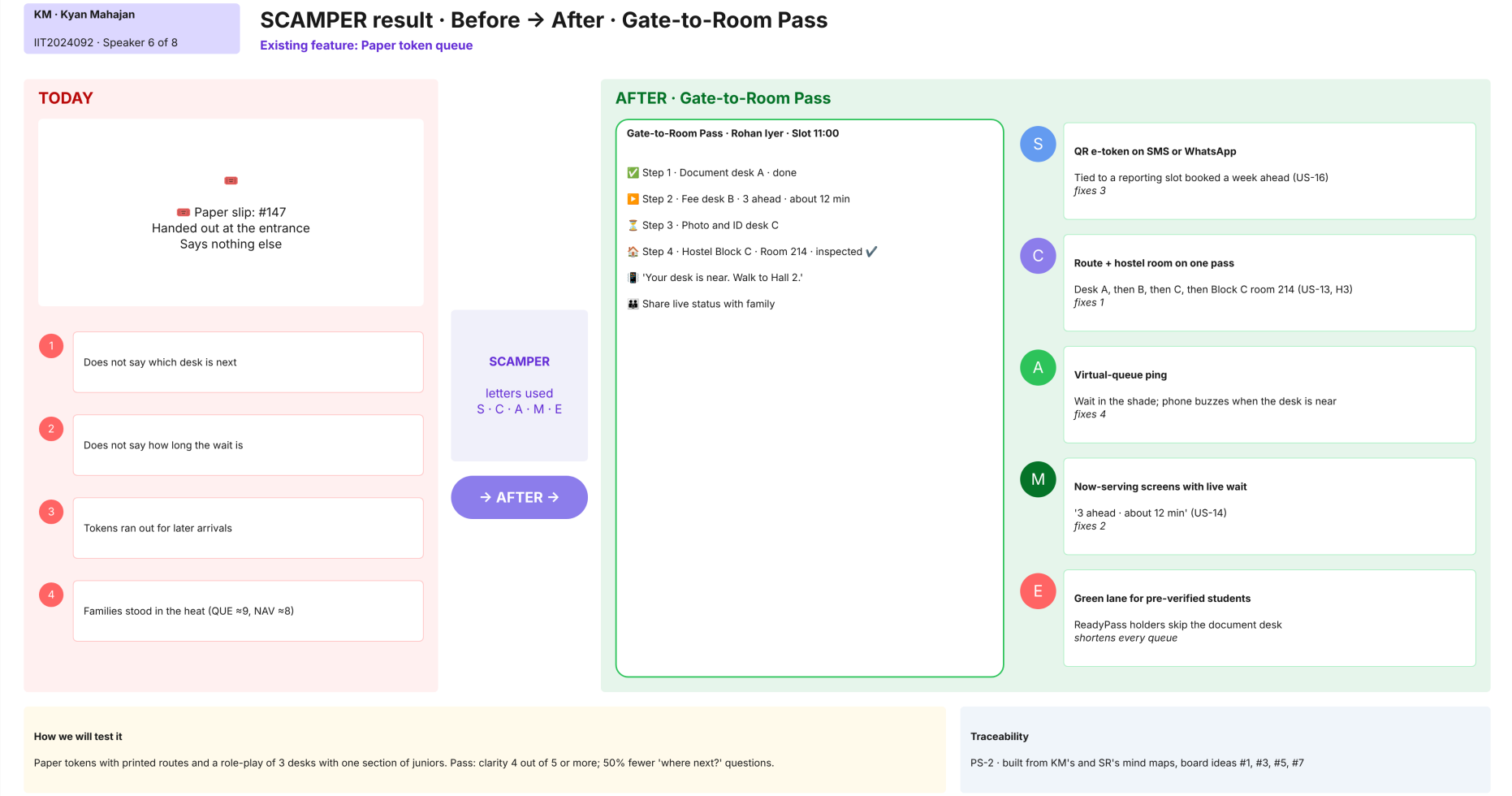
Each row is one letter with the variation, who benefits, and two ratings out of 5 (impact and feasibility). A gold star marks the rows we carried into the concept.

WHAT IT TELLS US

Ratings differ row by row; the stars mark the letters we kept. Result: Gate-to-Room Pass.

SCAMPER result · before → after · Gate-to-Room Pass

A numbered paper slip handed out at the entrance on reporting day.



WHAT IT IS

The left side is today's feature with its four pain points; the right side is the concept built from the letters we picked.

HOW TO READ IT

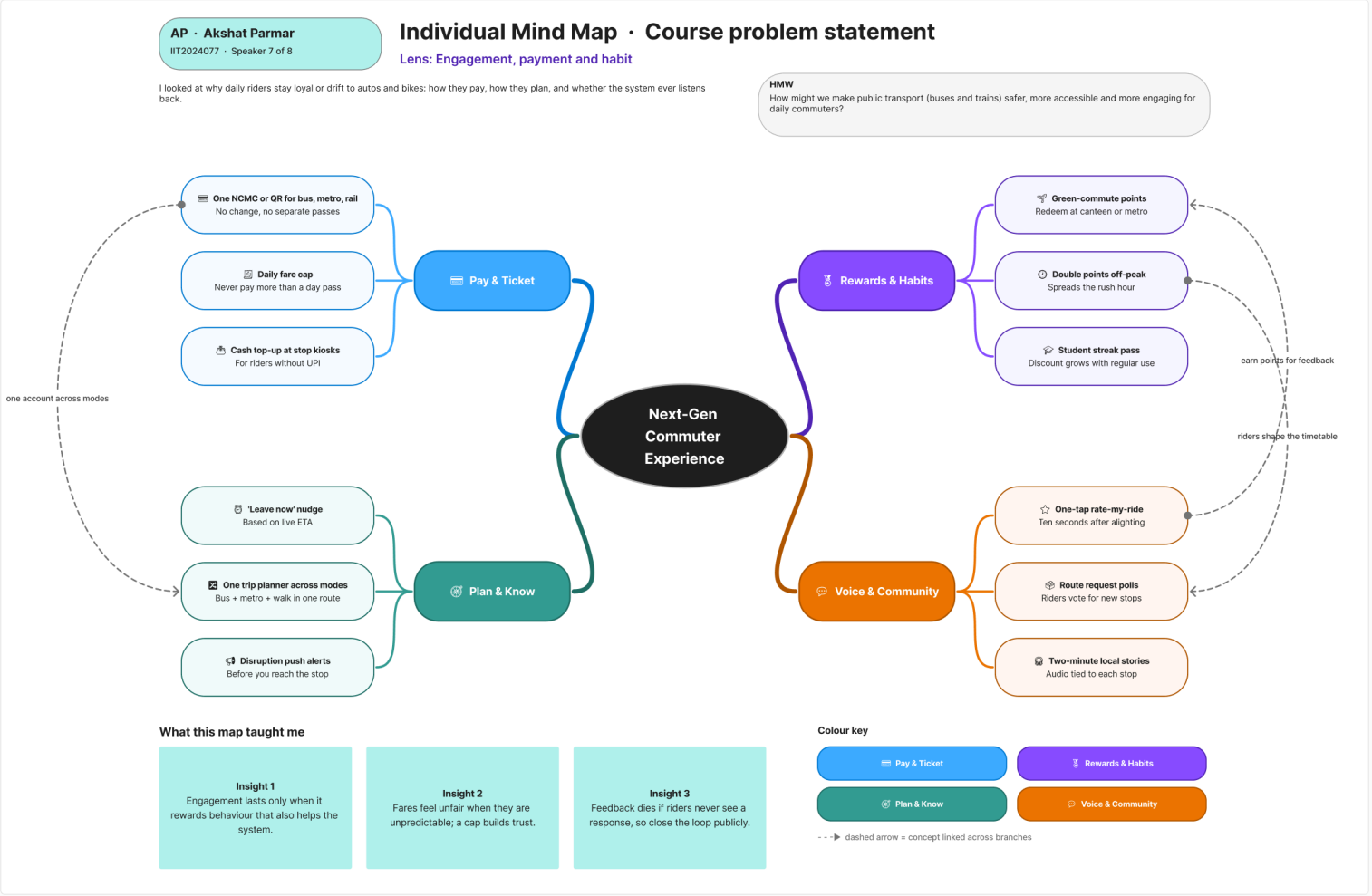
Each coloured circle on the right is the SCAMPER letter that produced that part; the 'fixes' note says which pain it removes. Bottom: how we will test it.

WHAT IT TELLS US

Gate-to-Room Pass answers every pain on the left and comes with a cheap prototype test and a pass mark.

Individual mind map · PS-T · Engagement, payment and habit

How might we make public transport (buses and trains) safer, more accessible and more engaging for daily commuters?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. I looked at why daily riders stay loyal or drift to autos and bikes: how they pay, how they plan, and whether the system ever listens back.

HOW TO READ IT

Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

Engagement lasts only when it rewards behaviour that also helps the system. · Fares feel unfair when they are unpredictable; a cap builds trust. · Feedback dies if riders never see a response, so close the loop publicly.

Converging · Course problem

Dot vote, impact × effort grid, weighted score

AP · Akshat Parmar

IIT2024077 · Speaker 7 of 8

Converging · Course problem

Dot vote → impact × effort → weighted score

Step 1 · Dot vote (8 members × 5 dots)

#1 Panic button on every seat row that alerts the driver and the nearest police van	7
#8 Handrail vibrates one stop before your destination	6
#7 Wheelchair users pre-book ramp and priority space via SMS	6
#3 Live crowd-density meter at stops, so you can skip a packed bus	5
#17 Seat-availability lights outside each coach	4
#11 Missed-call IVR gives bus ETA to feature-phone users	4
#13 Green-commute points redeemable at metro and canteen	3
#18 Single QR or NCMC card for bus, metro and rail	3
#4 Night request to get off between stops after 9 pm	2

Step 2 · Impact × effort

Quick wins

- LED stop panel (SCAMPER M)
- Missed-call IVR ETA (#11)
- Rate-my-ride (#16)

Big bets

- Smart Stop Strip modes (SCAMPER C + R)
- SOS + cabin clip (#1)
- Live crowd meter (#3)

Fill-ins

- Local-history audio (#14)

Avoid for now

- Auto-stop from ticket (SCAMPER E)

← low effort high effort → (up = more impact)

Step 3 · Weighted score

Concept	POV 30%	Evid. 25%	Feas. 20%	Hyp. 15%	Nov. 10%	Score
Smart Stop Strip ★	4	4	3	n/a	5	**3.9**
Night Safe-Ride kit (SOS + share link + request stop)	4	4	3	n/a	3	3.6
Inclusive info layer (IVR + multilingual + tactile)	4	3	4	n/a	3	3.6

Decision

Smart Stop Strip wins. The runners-up are not dropped: their best parts are folded into it.

WHAT IT IS

Convergence means choosing. Step 1 counts each member's dots on this board's ideas. Step 2 sorts the top ideas by impact and effort. Step 3 scores whole concepts on five weighted criteria.

HOW TO READ IT

Red dots are votes. In the grid, top-left is quick wins and top-right is big bets. In the table the green row is the winner; the weights in the column heads add up to 100%.

WHAT IT TELLS US

Smart Stop Strip (3.9 of 5) wins. The runners-up are not dropped: their best parts are folded into the winner.

Converging · PS-1 Certainty before travel

Dot vote, impact × effort grid, weighted score

AP · Akshat Parmar

IIT2024077 · Speaker 7 of 8

Converging · PS-1 Certainty before travel

Dot vote → impact × effort → weighted score

Step 1 · Dot vote (8 members × 5 dots)

#1 Personalised checklist generated from category, state and quota	7
#12 Downloadable 'Ready to Report' pass with QR, like airline web check-in	7
#10 Pull documents directly from DigiLocker or JoSAA	6
#7 Autosave every field; resume from the exact failed step	5
#13 Official WhatsApp FAQ bot that uses only the latest advisory	5
#2 Advisory frozen 10 days before reporting, with a visible version number	4
#11 Fee status auto-updates from the bank reference within 15 min	3
#4 'Notary needed?' tag on every document	3

Step 2 · Impact × effort

Quick wins

- Annotated samples (SCAMPER M)

Big bets

- Checklist = upload slots
- Ready to Report pass (#12)
- DigiLocker pull (#10)
- WhatsApp FAQ bot + seniors (#13)

Fill-ins

- Change-log page (#17)

Avoid for now

- Video-call document check (#14)

← low effort high effort → (up = more impact)

Step 3 · Weighted score

Concept	POV 30%	Evid. 25%	Feas. 20%	Hyp. 15%	Nov. 10%	Score
ReadyPass ★	5	5	3	5	4	**4.5**
Document Coach (samples + notary tags + video check)	4	5	5	5	2	4.4
Verified Helper Network (bot + seniors + ticket)	4	5	4	5	3	4.3

Decision

ReadyPass wins. The runners-up are not dropped: their best parts are folded into it.

WHAT IT IS

Convergence means choosing. Step 1 counts each member's dots on this board's ideas. Step 2 sorts the top ideas by impact and effort. Step 3 scores whole concepts on five weighted criteria.

HOW TO READ IT

Red dots are votes. In the grid, top-left is quick wins and top-right is big bets. In the table the green row is the winner; the weights in the column heads add up to 100%.

WHAT IT TELLS US

ReadyPass (4.5 of 5) wins. The runners-up are not dropped: their best parts are folded into the winner.

Converging · PS-2 Gate to a ready room

Dot vote, impact × effort grid, weighted score

AP · Akshat Parmar

IIT2024077 · Speaker 7 of 8

Converging · PS-2 Gate to a ready room

Dot vote → impact × effort → weighted score

Step 1 · Dot vote (8 members × 5 dots)



Step 2 · Impact × effort

Quick wins

- Now-serving screens (#7)
- Senior buddies
- Room inspection card (#12)

Big bets

- Route on e-token (#1)
- QR defect report (#13)
- Virtual-queue pings

Fill-ins

- Welcome film (#9)

Avoid for now

- Roaming verifiers (#10)

← low effort high effort → (up = more impact)

Step 3 · Weighted score

Concept	POV 30%	Evid. 25%	Feas. 20%	Hyp. 15%	Nov. 10%	Score
Gate-to-Room Pass ★	5	5	4	5	4	**4.7**
Room-Ready Guarantee (inspection + QR + kit)	4	5	4	5	3	4.3
Family Comfort Zone (lounge + seating + parent guide)	3	4	5	3	2	3.5

Decision

Gate-to-Room Pass wins. The runners-up are not dropped: their best parts are folded into it.

WHAT IT IS

Convergence means choosing. Step 1 counts each member's dots on this board's ideas. Step 2 sorts the top ideas by impact and effort. Step 3 scores whole concepts on five weighted criteria.

HOW TO READ IT

Red dots are votes. In the grid, top-left is quick wins and top-right is big bets. In the table the green row is the winner; the weights in the column heads add up to 100%.

WHAT IT TELLS US

Gate-to-Room Pass (4.7 of 5) wins. The runners-up are not dropped: their best parts are folded into the winner.

5.1 Selected concepts

ReadyPass · PS-1 Certainty before travel

A personalised checklist built from the student's category, in which **every line is an upload slot with a status light**. Items already verified through JoSAA or DigiLocker turn green automatically. Each item shows an annotated sample and a notarisation tag. When all items are green, the student and parents receive a **"Ready to Report" pass** with a QR code and reporting slot. Any rule change triggers an alert and a change-log entry.

Built from: SCAMPER S, C, A, E, R · Board ideas 1, 4, 7, 10, 12 · Mind-map links "prevent rejections" and "auto-accepted items" · **Tests:** H1, H2

Gate-to-Room Pass · PS-2 From gate to a ready room

A slot-based e-token that shows the student's **personal desk route, live wait time and hostel room**. It pings families when their desk is near, so they can wait in shaded seating. Pre-verified students use a green lane. **Badged senior buddies** answer questions from the same official FAQ. Every room passes an **inspection card** before handover and carries a **scan-to-report QR** with visible repair status.

Built from: SCAMPER S, C, A, M, E · Board ideas 1, 5, 7, 12, 13 · Mind-map links "same token data" and "buddy escalates" · **Tests:** H3, H4, H5

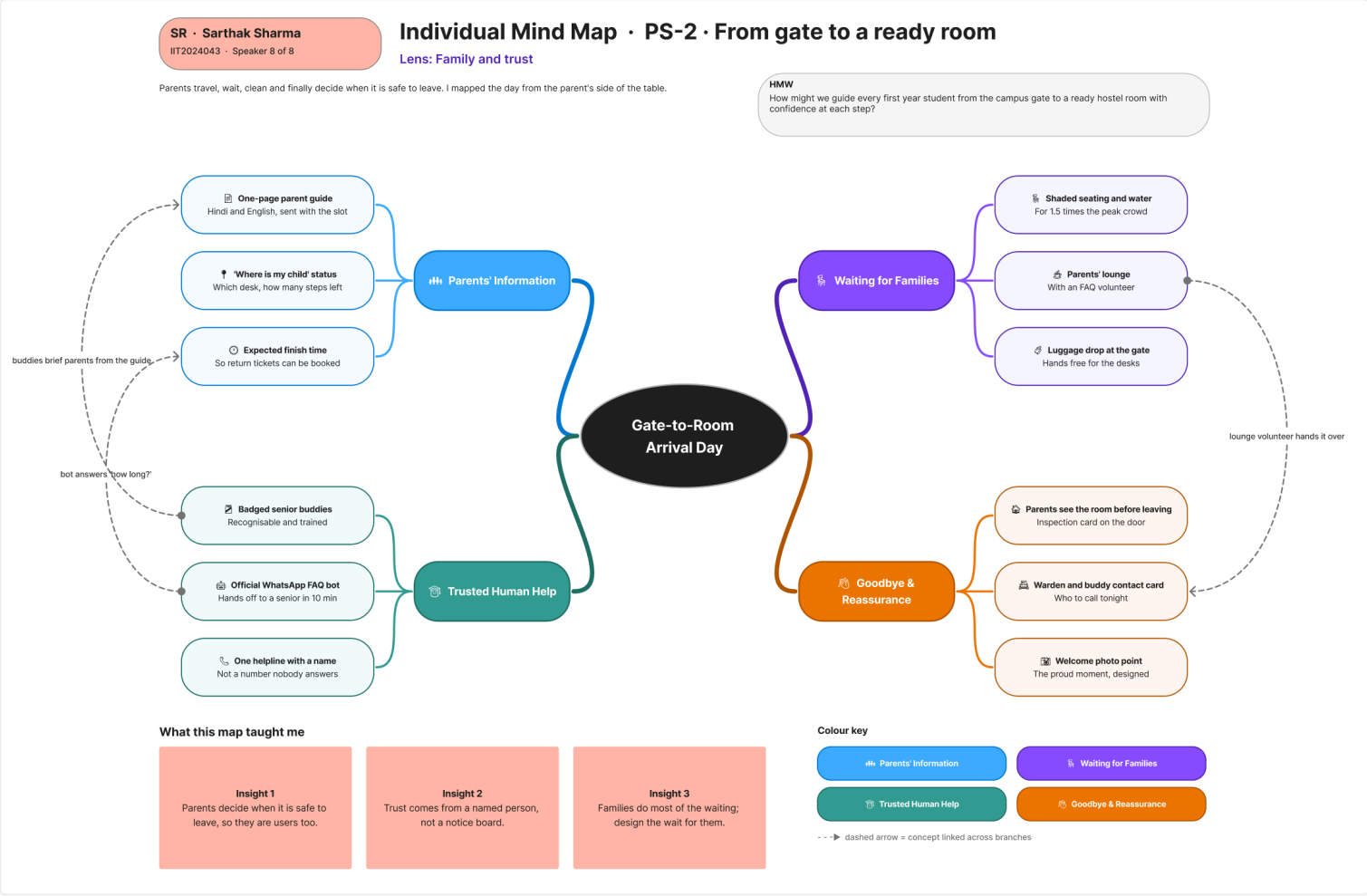
Smart Stop Strip · PS-T Next-gen commuter experience

A **touch strip along handrails** with three modes: tap to stop, hold for assistance (ramp or more time), triple-tap for a silent SOS to the driver and control room. A large LED panel confirms each request, and the passenger's ticket vibrates **"your stop is next"** one stop ahead.

Built from: SCAMPER S, C, A, M, R · Board ideas 1, 7, 8

Individual mind map · PS-2 · Family and trust

How might we guide every first year student from the campus gate to a ready hostel room with confidence at each step?



WHAT IT IS

A mind map is a drawing of everything one person can think of about a problem, spreading from one centre. Parents travel, wait, clean and finally decide when it is safe to leave. I mapped the day from the parent's side of the table.

HOW TO READ IT

Start at the black centre. Each coloured branch is one theme and each box on it is a specific idea. Dashed arrows join ideas from different branches that help each other. The three yellow notes at the bottom are what the map taught its author.

WHAT IT TELLS US

Parents decide when it is safe to leave, so they are users too. · Trust comes from a named person, not a notice board. · Families do most of the waiting; design the wait for them.

3.1 "Yes, and…" builds across the three boards

Each build kept the original idea and pushed it further. The builds were usually where the more novel ideas came from.

Board	Original idea	"Yes, and…" build	Cluster
PS-T	**SD**: Panic button on every seat row that alerts the driver and the nearest police van	**VS**: …and it also shares live GPS and a 10-second cabin clip	Feel Safe, Day & Night
PS-T	**SR**: Wheelchair users pre-book ramp and priority space via SMS	**SD**: …and the driver's dashboard shows which stop needs the ramp	Ride for Everyone
PS-T	**ST**: Green-commute points redeemable at metro and canteen	**SR**: …and double points for off-peak travel, to spread the crowd	Make the Ride Worth It
PS-T	**KM**: Seat-availability lights outside each coach	**VR**: …and the same data feeds the crowd meter at stops	Make the Ride Worth It
PS-1	**VS**: Personalised checklist generated from category, state and quota	**SS**: …and each item expands to a photo of a correct sample	Know Exactly What to Bring
PS-1	**VR**: Autosave every field; resume from the exact failed step	**KM**: …and an offline mode queues uploads until the hotspot reconnects	Upload Once, Confidently
PS-1	**KM**: Official WhatsApp FAQ bot that uses only the latest advisory	**VS**: …and anything it cannot answer goes to a senior volunteer within 10 min	Help That Answers Fast
PS-2	**SR**: Token shows a personal route: Desk A, then B, then C, then Hostel	**VR**: …and it pings your phone when your desk is ready, so you can sit anywhere	Move Without Getting Lost
PS-2	**SS**: Live now-serving screens per desk	**AP**: …and show the average time per student so families can plan	Wait Less, Wait Better
PS-2	**VS**: Pre-handover inspection card signed by the caretaker	**SR**: …and a photo of the inspected room is sent before arrival	A Room Ready to Live In

3.2 What clustering revealed

Board	Clusters (named by user need)	Observation
PS-T	Feel Safe, Day & Night · Ride for Everyone · Make the Ride Worth It	Safety ideas were the most numerous; accessibility ideas were the most specific. Engagement ideas worked best when they also helped the system run better.
PS-1	Know Exactly What to Bring · Upload Once, Confidently · Help That Answers Fast	The checklist cluster and the upload cluster kept merging, which showed the checklist and the upload should be one object (later SCAMPER "Combine").
PS-2	Move Without Getting Lost · Wait Less, Wait Better · A Room Ready to Live In	The token idea connected all three clusters, which made it the backbone of the PS-2 concept.

Storyboard: ReadyPass · Ananya Verma, Patna

Six frames of a user going through the concept

SR · Sarthak Sharma

IIT2024043 · Speaker 8 of 8

Storyboard · ReadyPass · Ananya Verma, Patna

PS-1 · six frames; the bar under each frame is how the user feels (1 anxious to 5 delighted)

1	2	3	4	5	6
					
10 days before SMS: 'Your personal checklist v3 is ready.'	**Opens it** 7 of 9 items already green from JoSAA and DigiLocker	**Affidavit** Downloads the template, sees the annotated sample, visits the notary once	**Next day** All 9 green. Ready to Report pass with slot 10:30 arrives. Father books the return train.	**A rule changes** Alert and change-log entry reach her before travel, not at the desk	**Reporting day** Shows the QR, takes the green lane, done by 11:15. Father catches his train.
feels: curious ●●●○○	feels: relieved ●●●●○	feels: confident ●●●●○	feels: happy ●●●●●	feels: calm ●●●●○	feels: proud ●●●●●

WHAT IT IS

A storyboard is a comic strip of one user going through a concept. We use it to check the idea makes sense before building anything.

HOW TO READ IT

Read left to right. The emoji is the scene, the caption says what happens, and the coloured bar is how the user feels (1 anxious to 5 delighted).

WHAT IT TELLS US

The mood climbs from uncertain to settled; the lowest frame is the first thing our test plan checks.

Storyboard: Gate-to-Room Pass · Rohan Iyer and Meera Nair

Six frames of a user going through the concept

SR · Sarthak Sharma

IIT2024043 · Speaker 8 of 8

Storyboard · Gate-to-Room Pass · Rohan Iyer and Meera Nair

PS-2 · six frames; the bar under each frame is how the user feels (1 anxious to 5 delighted)

1	2	3	4	5	6
					
At the gate Scans the pass; the screen shows his 4-step route	**Waiting** Mother sits in the shaded lounge; phone buzzes when desk B is near	**Desk B** Screen says '3 ahead · about 12 min'. No guessing.	**Senior buddy** A badged buddy walks the family to Block C	**Room 214** Signed inspection card on the door; fan, light and lock checked	**One loose latch** Meera scans the door QR; repair status shows 'fixed in 2 h'. Parents leave calm.
feels: oriented ●●●○○	feels: rested ●●●●○	feels: patient ●●●○○	feels: welcomed ●●●●○	feels: reassured ●●●●●	feels: settled ●●●●●

WHAT IT IS

A storyboard is a comic strip of one user going through a concept. We use it to check the idea makes sense before building anything.

HOW TO READ IT

Read left to right. The emoji is the scene, the caption says what happens, and the coloured bar is how the user feels (1 anxious to 5 delighted).

WHAT IT TELLS US

The family is calm by the end because the room and the status are visible.

Storyboard: Smart Stop Strip · daily commuter

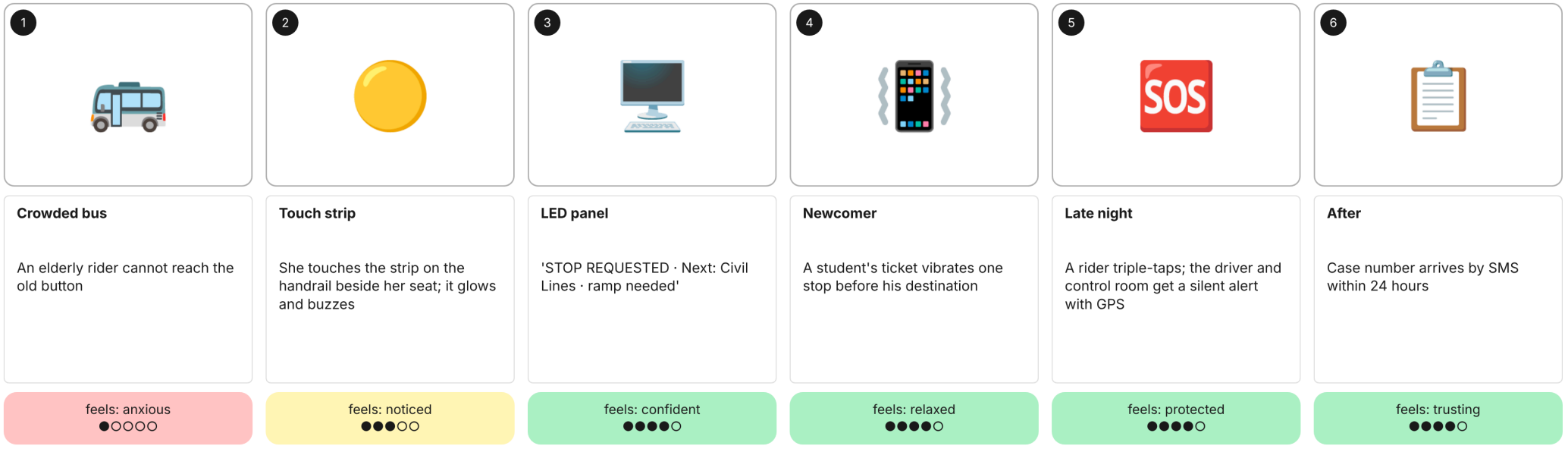
Six frames of a user going through the concept

SR · Sarthak Sharma

IIT2024043 · Speaker 8 of 8

Storyboard · Smart Stop Strip · daily commuter

PS-T · six frames; the bar under each frame is how the user feels (1 anxious to 5 delighted)



WHAT IT IS

A storyboard is a comic strip of one user going through a concept. We use it to check the idea makes sense before building anything.

HOW TO READ IT

Read left to right. The emoji is the scene, the caption says what happens, and the coloured bar is how the user feels (1 anxious to 5 delighted).

WHAT IT TELLS US

The riskiest frame is the late-night SOS, so it gets the first prototype test.

5.2 Next step: prototype and test

Concept	Low-fidelity prototype	Test with	Success signal (from A2 hypotheses)
ReadyPass	Clickable Figma flow: checklist, status lights, Ready pass	10 to 15 current first years plus 3 parents	85% or more identify all documents; 80% or more say they would feel ready to travel
Gate-to-Room Pass	Paper tokens with printed routes; role-play of 3 desks; room inspection card on 5 rooms	One section of juniors acting as arrivals, plus 2 wardens	Clarity rating 4 out of 5 or higher; 50% fewer “where next?” questions; 90% of rooms pass
Smart Stop Strip	Cardboard handrail with an LED strip; storyboard of the three modes	8 commuters, including 2 elderly and 1 wheelchair user	All participants trigger the correct mode on the first try

Generative AI acknowledgement

Claude (Anthropic) was used as an assistant, October 2026. The thinking and design work is ours.

WHERE AI ASSISTED

- Suggesting icons for mind-map labels
- A few extra SCAMPER variations to compare against our own
- Checking that ratings match the reasons we wrote
- Help laying out boards and tables

WHAT WAS OUR OWN WORK

- All research evidence (our own 30 Empathize records)
- Every insight, persona, problem statement, hypothesis, idea, rating and decision
- The mind maps, brainstorm sessions and SCAMPER work, done by the named members
- Final wording and checking of every page

Purpose	Prompts we used (examples)
Icons	Suggest one icon or emoji for each of these mind-map labels.
SCAMPER	Here is our variation for Combine. Suggest two other ways to word it.
Rating check	Do any of these 1 to 5 ratings look inconsistent with the reason I wrote?
Consistency	Check that idea numbers on the board match the ones in the concept summary.
Layout	Help me fit this scoring table on one page.

How AI was used, member by member

Tool: Claude (Anthropic), October 2026. The evidence is our own Empathize research (30 student records). The team declaration follows.

Member	Section	Where AI helped	What we did ourselves
Saumya Sood IIT2024018	PS-T accessibility map, PS-T board	Suggested icons for map nodes.	Thought through the accessibility branches, facilitated the PS-T brainstorm.
Sudhanshu IIT2024081	PS-T safety map, SCAMPER stop button	Suggested a few SCAMPER variations to compare.	Chose the safety branches, rated the stop-button variations, picked the letters.
Vishwas IIT2024087	PS-1 ERP map, PS-1 board	Suggested icons for map nodes.	Built the ERP and upload branches, facilitated the PS-1 brainstorm.
Vichanshu Raj IIT2024083	PS-1 documents map, SCAMPER advisory PDF	Suggested a few SCAMPER variations to compare.	Built the document branches, rated the advisory PDF variations.
Satwik Santosh IIT2024084	PS-2 hostel-room map, PS-2 board	Suggested icons for map nodes.	Built the hostel-room branches, facilitated the PS-2 brainstorm.
Kyan Mahajan IIT2024092	PS-2 queue map, SCAMPER paper token	Suggested a few SCAMPER variations to compare.	Built the queue and wayfinding branches, rated the paper token variations.
Akshat Parmar IIT2024077	PS-T engagement map, convergence	Helped format the scoring table.	Built the engagement branches, counted the dots, set the scoring weights.
Sarthak Sharma IIT2024043	PS-2 family-trust map, storyboards, test plan	Helped lay out the storyboard frames.	Built the family and trust branches, wrote the storyboards and test plan.